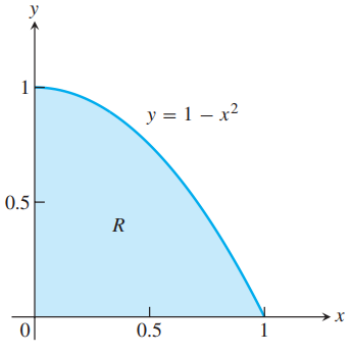
Integration

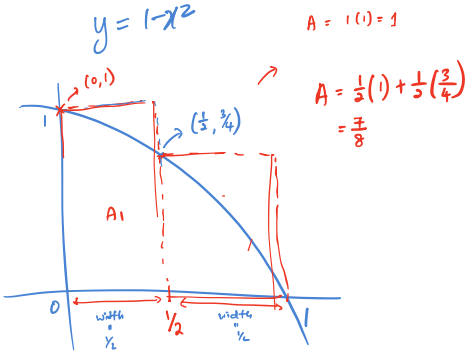
In this chapter we study a method to calculate the areas and volumes of these and other more general shapes. The method we develop, called integration, is a tool for calculating much more than areas and volumes. The integral has many applications in statistics, economics, the sciences, and engineering.

Estimating with Finite Sums

The area of a region with a curved boundary can be approximated by summing the areas of a collection of rectangles. Using more rectangles can increase the accuracy of the approximation.

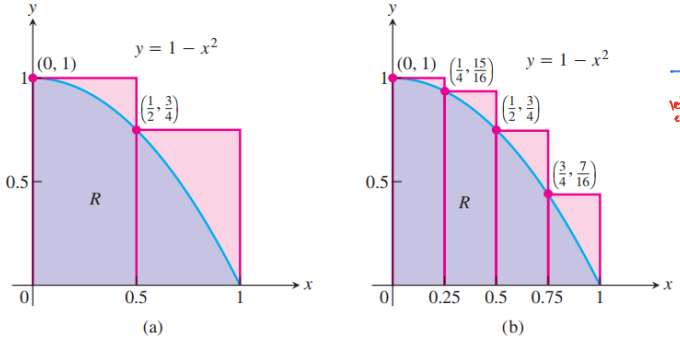
Example. *Approximating Area.*

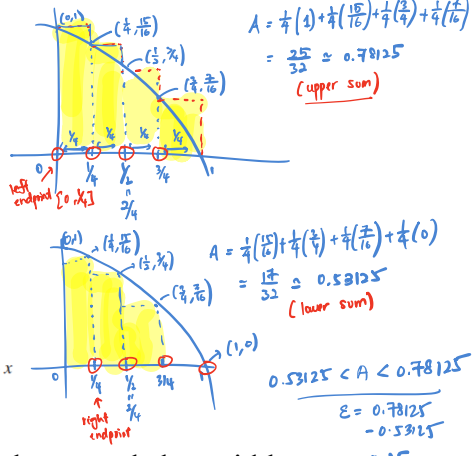




What is the area of the shaded region R that lies above x-axis, below the graph of $y = 1 - x^2$, and between the vertical lines $x = 0$ and $x = 1$? Unfortunately, there is no simple geometric formula for calculating the areas of shapes having curved boundaries like the region R.

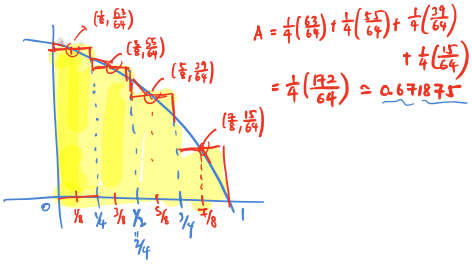
While we do not yet have a method for determining the exact area of R, we can approximate it in a simple way.





In Figure (a) shows two rectangles that together contain the region R. Each rectangle has width $1/2$ and they have heights 1 and ¾, moving from left to right. The height of each rectangle is the maximum value of the function f, obtained by evaluating f at the left endpoint of the subinterval of $[0,1]$ forming the base of the rectangle. The total area of these tow rectangles approximates the area A of the region R. This estimation is *larger* than the true area of A.

$$A \approx 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{8} = 0.875.$$



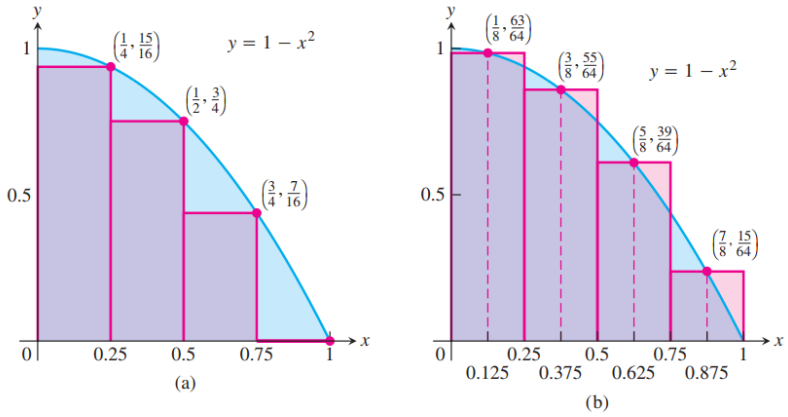
We say 0.875 is an ***upper sum*** because it obtained by taking the height of each rectangle as the maximum (uppermost) value of $f(x)$ for x a point in the base interval of the rectangle.

In Figure (b), we improve our estimation by using four thinner rectangles, each of width $1/4$, which taken together contain the region R. These four rectangles give the approximation

$$A \approx 1\left(\frac{1}{4}\right) + \frac{15}{16}\left(\frac{1}{4}\right) + \frac{3}{4}\left(\frac{1}{4}\right) + \frac{7}{16}\left(\frac{1}{4}\right) = \frac{25}{32} = 0.78125,$$

which is still greater than A since the four rectangles contains R.

Suppose instead we use four rectangles contained *inside* the region R to estimate the area.



By looking at (a) in figure above, each rectangle has width $1/4$ as before, but the rectangles are shorter and lie entirely beneath the graph of f. The function $f(x) = 1 - x^2$ is decreasing on $[0,1]$, so the height of each of these rectangles is given by the value of f at the right endpoint of the subinterval forming its base. The fourth rectangle has zero height and therefore contribute no area. Summing these rectangles with heights equal to the minimum value of $f(x)$ for x a point in each base subinterval, gives a ***lower sum*** approximation to the area.

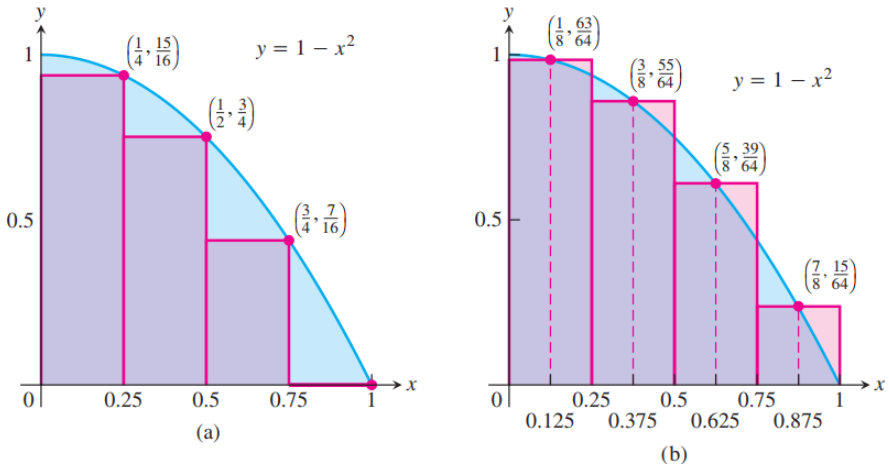
$$A \approx \frac{15}{16}\left(\frac{1}{4}\right) + \frac{3}{4}\left(\frac{1}{4}\right) + \frac{7}{16}\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) = \frac{17}{32} = 0.53125.$$

This estimation is smaller than the area A since the rectangle all lie inside of the region R. The true value of A lies somewhere between these lower and upper sums.

$$0.53125 < A < 0.78125.$$

By considering both lower and upper sum approximations we get not only estimates for the area, but also a bound on the size of the possible error in these estimations since the true value of the area lies somewhere between them. Here the error cannot be greater than the difference $0.78125 - 0.53125 = 0.25$.

Another estimate can be obtained by using rectangles whose heights are the values of f at the *midpoints* of their bases. This method of estimation is called the ***midpoint rule*** for approximating the area. The midpoint rule gives an estimate that is between a lower sum and an upper sum, but it is not clear whether it overestimates or underestimates the true area.



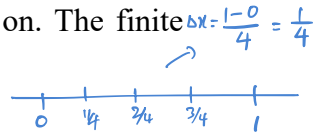
With four rectangles of width $1/4$ as before, the midpoint rule estimates the area of R to be

$$A \approx \frac{63}{64}\left(\frac{1}{4}\right) + \frac{55}{64}\left(\frac{1}{4}\right) + \frac{39}{64}\left(\frac{1}{4}\right) + \frac{15}{64}\left(\frac{1}{4}\right) = \frac{172}{64}\left(\frac{1}{4}\right) = 0.671875.$$

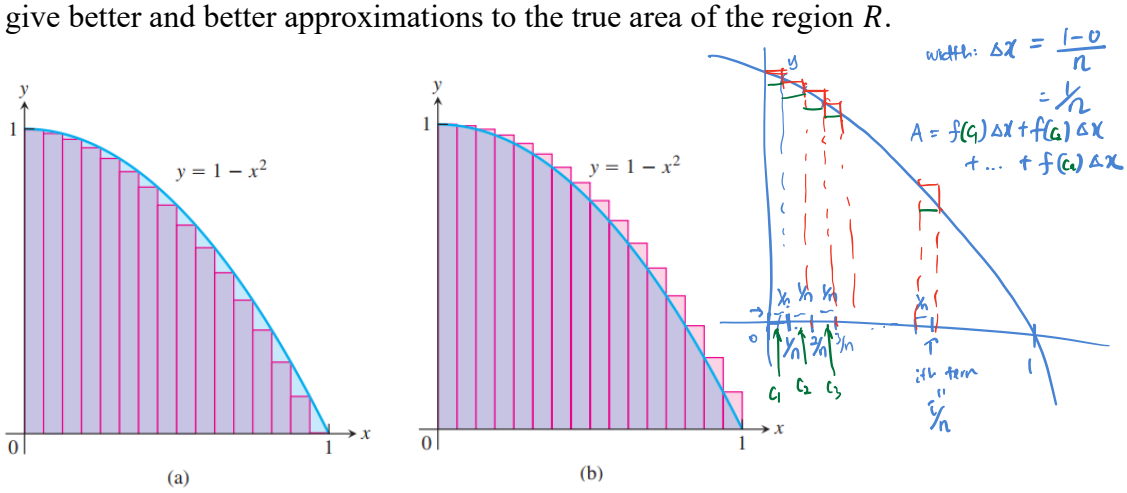
In each of our computed sums, the interval $[a, b]$ over which the function f is defined was subdivided into n subintervals of equal width (also called length) $\Delta x = (b - a)/n$, and f was evaluated at a point in each subinterval:

c_1 in first subinterval, c_2 in second subinterval, c_3 in third subinterval and so on. The finite sums then all take the form

$$f(c_1)\Delta x + f(c_2)\Delta x + f(c_3)\Delta x + \cdots + f(c_n)\Delta x.$$



By taking more and more rectangle, with each rectangle thinner than before, it appears that these finite sums give better and better approximations to the true area of the region R.



In the Figure (a) shows that a lower sum approximation for the area of R using 16 rectangles of equal width. The sum of their areas is 0.634765625, which appears close to the true area, but still smaller since the rectangles lie inside R.

The Figure (b) shows that a upper sum approximation using 16 rectangles of equal width. The sum of their areas is 0.697265625, which is somehow larger than the true area because the rectangles taken together contain R. The midpoint rule for 16 rectangles gives a

total area approximation of 0.6669921875, but it is not immediately clear whether this estimate is larger or smaller than the true area.

Finite approximations for the area of R

Number of subintervals	Lower sum	Midpoint rule	Upper sum
2	.375	.6875	.875
4	.53125	.671875	.78125
16	.634765625	.6669921875	.697265625
50	.6566	.6667	.6766
100	.66165	.666675	.67165
1000	.6661665	.66666675	.6671665

Table above shows the values of upper and lower sum approximations to the area of R using up to 1000 rectangles. From the values in the table, it looks as if R is approaching $2/3$ as numbers of rectangle increases.

Average Value of a Nonnegative Continuous Function

The average value of a collection of n numbers $x_1, x_2, \ldots, x_n$ is obtained by adding them together and dividing by n.

What is the average value of a continuous function f on an interval $[a, b]$?

Such a function can assume infinitely many values.

For example, the temperature at a certain location in a town is a continuous function that goes up and down each day. What does it mean to say that the average temperature in the town over the course of a day is 73 degrees?

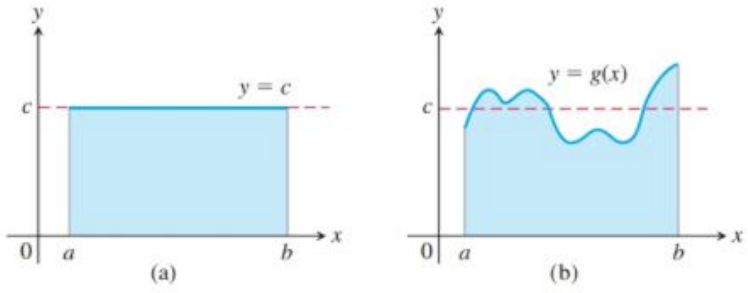
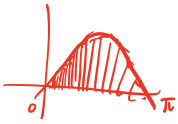


FIGURE 5.6 (a) The average value of $f(x) = c$ on $[a, b]$ is the area of the rectangle divided by $b - a$. (b) The average value of $g(x)$ on $[a, b]$ is the area beneath its graph divided by $b - a$.

When a function is constant, this question is easy to answer. A function with constant value c on an interval $[a, b]$ has average value c. When c is positive, its graph over $[a, b]$ gives a rectangle of height c. The average value of the function can then be interpreted geometrically as the area of this rectangle divided by its width $b - a$.

What if we want to find the average value of a nonconstant function, such as the function g in Figure 5.6b?

We are led to define the average value of a nonnegative function on an interval $[a, b]$ to be the area under its graph divided by $b - a$. For this definition to be valid, we need a precise understanding of what is meant by the area under a graph. For now we look at an example.



Example. Estimate the average value of the function $f(x) = \sin x$ on the interval $[0, \pi]$.

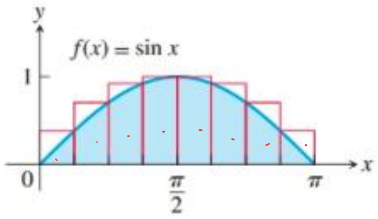


FIGURE 5.7 Approximating the area under $f(x) = \sin x$ between 0 and π to compute the average value of $\sin x$ over $[0, \pi]$, using eight rectangles (Example 3).

To find the average, we need to calculate the area A under the graph and then divide this area by the length of the interval, $\pi - 0 = \pi$.

We do not have a simple way to determine the area, so we approximate it with finite sums.

To get an upper sum approximation,

$$A \approx \left(\sin\frac{\pi}{8} + \sin\frac{\pi}{4} + \sin\frac{3\pi}{8} + \sin\frac{\pi}{2} + \sin\frac{\pi}{2} + \sin\frac{5\pi}{8} + \sin\frac{3\pi}{4} + \sin\frac{7\pi}{8}\right)\left(\frac{\pi}{8}\right)$$

$$\approx 2.364$$

Average: $\frac{2.364}{\pi} \approx 0.753$.



TABLE 5.5 Average value of $\sin x$ on $0 \le x \le \pi$

Number of subintervals	Upper sum estimate
8	0.75342
16	0.69707
32	0.65212
50	0.64657
100	0.64161
1000	0.63712

To estimate the average value of $\sin x$ on $[0, \pi]$ we divide the estimated area by the length π of the interval and obtain the approximation $\frac{2.364}{\pi} \approx 0.753$.

Since we used an upper sum to approximate the area, this estimate is greater than the actual average value of $\sin x$ over $[0, \pi]$. If we use more and more rectangles, with each rectangle getting thinner and thinner, we get closer and closer to the exact average value. (The true value is $\frac{2}{\pi} \approx 0.63662$)

Summary

The area under the graph of a positive function, the distance travelled by a moving object that doesn't change direction, and the average value of a nonnegative function f over an interval can all be approximated by finite sums constructed in a certain way.

First we sub-divide the interval into subintervals, treating f as if it were constant over each subinterval.

Then we multiply the width of each subinterval by the value of f at some point within it and add these products together.

If the interval $[a, b]$ is subdivided into n intervals of equal widths $\Delta x = (b - a)/n$, and if $f(c_k)$ is the value of f at the chosen point c_k in the k-th subinterval, this process gives a finite sum of the form

$$f(c_1)\Delta x + f(c_2)\Delta x + f(c_3)\Delta x + \cdots + f(c_n)\Delta x.$$

The choices of the c_k could maximize or minimize the value of f in the kth subinterval, or give some value in between.

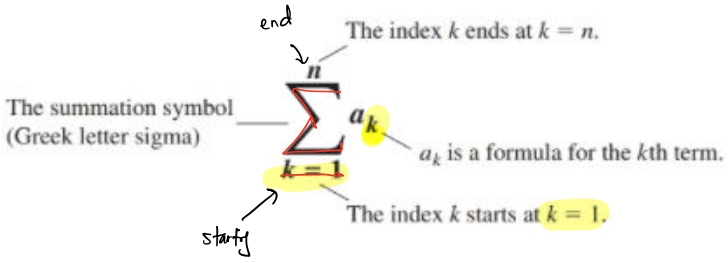
The finite sum approximations improved as we took more subintervals of smaller width.

Sigma Notation and Limits of Finite Sums

Finite Sums and Sigma Notation

Sigma notation enables us to write a sum with many terms in the compact form

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + \cdots + a_{n-1} + a_n.$$



Thus we can write the sum of the squares of the numbers 1 through 11 as

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2 + 11^2 = \sum_{k=1}^{11} k^2.$$

The sum of $f(i)$ for integers i from 1 to 100 as

$$f(1) + f(2) + f(3) + \cdots + f(100) = \sum_{i=1}^{100} f(i).$$

The starting index does not have to be 1; it can be any integer.

Example.

Express the sum 1 + 3 + 5 + 7 + 9 in sigma notation.

EXAMPLE 1

A sum in sigma notation	The sum written out, one term for each value of k	The value of the sum
$\sum_{k=1}^{5} k$	$1 + 2 + 3 + 4 + 5$	15
$\sum_{k=1}^{3} (-1)^k k$	$(-1)^1(1) + (-1)^2(2) + (-1)^3(3)$	$-1 + 2 - 3 = -2$
$\sum_{k=1}^{2} \frac{k}{k+1}$	$\frac{1}{1+1} + \frac{2}{2+1}$	$\frac{1}{2} + \frac{2}{3} = \frac{7}{6}$
$\sum_{k=4}^{5} \frac{k^2}{k-1}$	$\frac{4^2}{4-1} + \frac{5^2}{5-1}$	$\frac{16}{3} + \frac{25}{4} = \frac{139}{12}$

Note:

$$\sum_{k=1}^{3} (k + k^2),$$

We can rearrange into

$$\sum_{k=1}^{3} (k + k^2) = (1 + 1^2) + (2 + 2^2)(3 + 3^2)$$

$$= (1 + 2 + 3) + (1^2 + 2^2 + 3^2)$$

$$= \sum_{k=1}^{3} k + \sum_{k=1}^{3} k^2.$$

This illustrates a general rule for finite sums:

$$\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k.$$

Algebra Rules for Finite Sums

1. Sum Rule:
$$\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$$

2. Different Rule:
$$\sum_{k=1}^{n} (a_k - b_k) = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n} b_k$$

3. Constant Multiple Rule:
$$\sum_{k=1}^{n} ca_k = c \cdot \sum_{k=1}^{n} a_k$$

4. Constant Value Rule:
$$\sum_{k=1}^{n} c = n \cdot c$$

$\sum_{k=1}^{n}(3k-k^2) = \sum_{k=1}^{n}3k - \sum_{k=1}^{n}k^2 = 3\sum_{k=1}^{n}k - \sum_{k=1}^{n}k^2$

$\sum_{k=1}^{n}(-1)a_k = (-1)\sum_{k=1}^{n}a_k$

$\sum_{k=1}^{3}k+4 = \sum_{k=1}^{3}k + \sum_{k=1}^{3}4 = \sum_{k=1}^{3}k + 4\left(\sum_{k=1}^{3}1\right)$

Example.

(a) $\sum_{k=1}^{n}(3k-k^2) = 3\sum_{k=1}^{n}k - \sum_{k=1}^{n}k^2$

(b) $\sum_{k=1}^{n}(-a_k) = (-1)\sum_{k=1}^{n}a_k$

(c) $\sum_{k=1}^{3}(k+4) = \sum_{k=1}^{3}k + \sum_{k=1}^{3}4 = (1+2+3)+(4+4+4) = 6+12 = 18.$

$= (1+2+3) + 4(1+1+1)$

$= 6 + 4(3) = 18.$

(d) $\sum_{k=1}^{n}\frac{1}{n} = \frac{1}{n}\sum_{k=1}^{n}1 = \frac{1}{n}(n) = 1.$

a_k = formula in term of k.

$\sum_{k=1}^{n}\left(\frac{1}{n}\right) = \frac{1}{n}\sum_{k=1}^{n}1 = \frac{1}{n}(n) = 1.$

$a_k = 1/n$

Limit of Finite Sums

(a) The sum of the first n integers $\quad \sum_{k=1}^{n}k = \frac{n(n+1)}{2}$

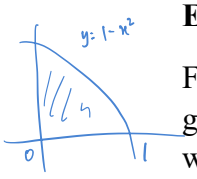
$1+2+3+\cdots+n = \sum_{k=1}^{n}k = \frac{n(n+1)}{2}$

(b) The sum of first n squares $\quad \sum_{k=1}^{n}k^2 = \frac{n(n+1)(2n+1)}{6}$

$1^2+2^2+3^2+\cdots+n^2 = \sum_{k=1}^{n}k^2 = \frac{n(n+1)(2n+1)}{6}$

(c) The sum of first n cubes $\quad \sum_{k=1}^{n}k^3 = \left(\frac{n(n+1)}{2}\right)^2.$

$1^3+2^3+3^3+\cdots+n^3 = \sum_{k=1}^{n}k^3 = \left(\frac{n(n+1)}{2}\right)^2$



Example. *The Limit of Finite Approximations to an Area*

Find the limiting value of lower sum approximations to the area of the region R below the graph of $y = 1-x^2$ and above the interval $[0,1]$ on the x-axis using equal width rectangles whose widths approach zero and whose number approaches infinity. $n \to \infty$

Solution. We compute a lower sum approximation using n rectangle of equal width $\Delta x = \frac{1-0}{n}$, and then we see what happens as $n \to \infty$. We start by subdividing $[0,1]$ into n equal width subintervals

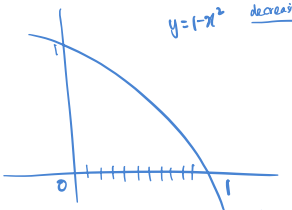
$$\left[0,\frac{1}{n}\right],\left[\frac{1}{n},\frac{2}{n}\right],\left[\frac{2}{n},\frac{3}{n}\right],\cdots,\left[\frac{n-1}{n},1\right].$$

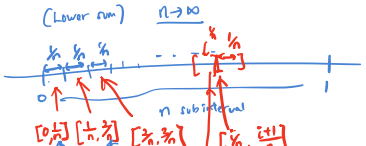
Each subinterval has width $\frac{1}{n}$. The function $1-x^2$ is decreasing on $[0,1]$, and its smallest value in a subinterval occurs at the subinterval's right endpoint. So a lower sum is constructed with rectangles whose height over the subinterval $\left[\frac{k-1}{n},\frac{k}{n}\right]$ is $f\left(\frac{k}{n}\right) = 1-\left(\frac{k}{n}\right)^2$, giving the sum

$$f\left(\frac{1}{n}\right)\cdot\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right)\cdot\left(\frac{1}{n}\right) + \cdots + f\left(\frac{k}{n}\right)\cdot\left(\frac{1}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\cdot\left(\frac{1}{n}\right).$$

$\Delta x = \frac{1-0}{n} = \frac{1}{n}$

We write this in sigma notation and simplify,





$A = \frac{1}{n}\left(f\left(\frac{1}{n}\right)\right) + \frac{1}{n}f\left(\frac{2}{n}\right) + \cdots + \frac{1}{n}f\left(\frac{k}{n}\right) + \cdots + \frac{1}{n}f\left(\frac{n}{n}\right).$

$= \frac{1}{n}\left(f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \cdots + f\left(\frac{k}{n}\right) + \cdots + f(1)\right)$

$$\sum_{k=1}^{n} f\left(\frac{k}{n}\right)\left(\frac{1}{n}\right) = \sum_{k=1}^{n}\left(1-\left(\frac{k}{n}\right)^2\right)\left(\frac{1}{n}\right)$$

$$= \sum_{k=1}^{n}\left(\frac{1}{n} - \frac{k^2}{n^3}\right)$$

$$= \sum_{k=1}^{n}\frac{1}{n} - \sum_{k=1}^{n}\frac{k^2}{n^3}$$

$$= n\left(\frac{1}{n}\right) - \frac{1}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right)$$

$$= 1 - \frac{1}{6}\left(1\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right).$$

We have obtained an expression for thew lower sum that holds for any n. Taking the limit of this expression as $n \to \infty$, we see that the lower sums converge as the number of subintervals increases and the subinterval widths approach zero:

$$\lim_{n\to\infty}\left(1 - \frac{1}{6}\left(1\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right)\right) = 1 - \frac{2}{6} = \frac{2}{3}.$$

The lower sum approximations converge to $2/3$. A similar calculation shows that the upper sum approximations also converge to $2/3$.

Riemann Sums

The theory of limits of finite approximations was made precise by the German mathematician Bernhard Riemann. We now introduce the notion of a *Riemann sum*, which underlies the theory of the definite integral that will be presented in the next section.

We begin with an arbitrary bounded function f defined on a closed interval $[a, b]$. Like the function pictured in Figure 5.8, f may have negative as well as positive values.

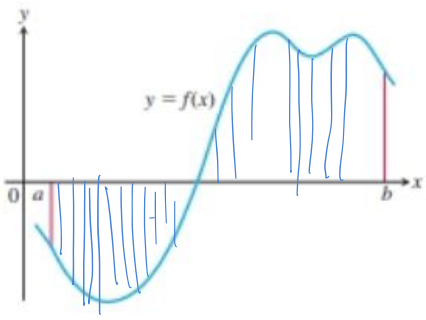


FIGURE 5.8 A typical continuous function $y = f(x)$ over a closed interval $[a, b]$.

We subdivide the interval $[a, b]$ into subintervals, not necessarily of equal width (or length), and form sums in the same way as for the finite approximations.
To do so, we choose $n - 1$ points $\{x_1, x_2, x_3, \ldots, x_{n-1}\}$ between a and b.

$$a = x_0 < x_1 < x_2 < x_3 < \cdots < x_{n-1} < b = x_n.$$

The set of all these points,

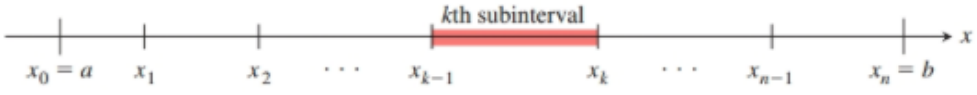
$$P = \{x_0, x_1, x_2, \ldots, x_{n-1}, x_n\}$$

Is called a **partition** of $[a, b]$.

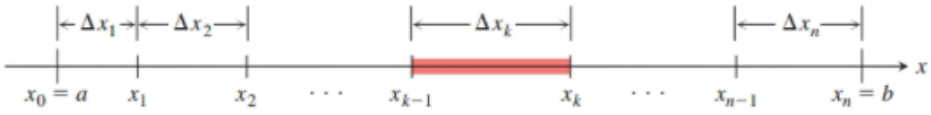
The partition P divides $[a, b]$ into n closed subintervals

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

The first of these subintervals is $[x_0, x_1]$, the second is $[x_1, x_2]$ and the k-th subinterval is $[x_{k-1}, x_k]$ (k an integer between 1 to n).



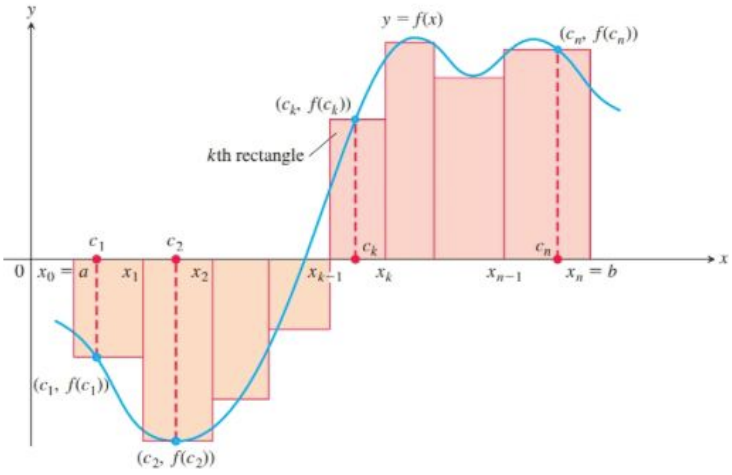
The width of the first subinterval $[x_0, x_1]$ is denoted Δx_1, the width of the second $[x_1, x_2]$ is Δx_2 and the width of the k-th subinterval is $\Delta x_k = x_k - x_{k-1}$.



If all n subintervals has equal width, the their common width, we call $\Delta x = \frac{b-a}{n}$.

Using equal width subintervals is often the simplest choice when doing computations.

The point chosen in the k-th subinterval $[x_{k-1}, x_k]$ is called c_k. Then on each subinterval, we stand a vertical rectangle that stretches from the x-axis to touch the curve at $(c_k, f(c_k))$.



On each subinterval we form the product $f(c_k)\Delta x_k$. This product is positive, negative, or zero, depending on the sign of $f(c_k)$.

Finally, we sum all these products to get

$$S_p = \sum_{k=1}^{n} f(c_k)\Delta x_k.$$

$S_p = \sum f(c_k)\,\Delta x_k$

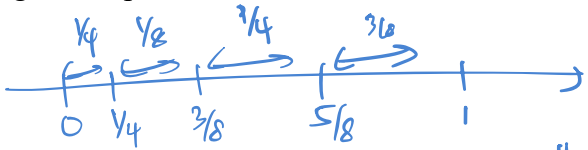
$P = \{x_0, x_1, \dots, x_n\}$

$[a, b]$

The sum S_P is called a **Riemann sum** for f on the interval $[a, b]$.

$\|P\|$ norm / length of P. (largest)

-There are many such sums, depending on the partition P we choose and on the choices of the points c_k in the subintervals.



$P = \{0, 1/4, 3/8, 5/8, 1\}$

$\|P\| = 3/8$

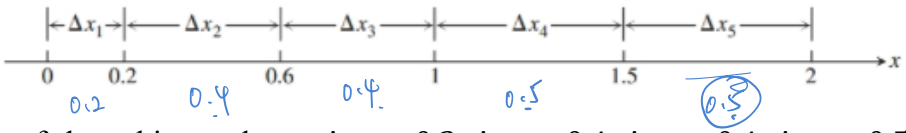
We could choose n subintervals all having equal width, $\Delta x = \frac{b-a}{n}$ to partition $[a, b]$. Choose c_k to be the right-hand endpoint of each subinterval.

$$S_n = \sum_{k=1}^{n} f\left(a + \frac{k(b-a)}{n}\right)\left(\frac{b-a}{n}\right).$$

we can make them thinner by simply increasing their number n. When a partition has subintervals of varying widths, we can ensure they are all thin by controlling the width of a widest (longest) subinterval.

We define the norm of a partition P, written $||P||$, to be the largest of all the subinterval widths. If P is a small number, then all of the subintervals in the partition P have a small width.

Example. The set $P = \{0, 0.2, 0.6, 1, 1.5, 2\}$ is a partition of $[0,2]$. There are five subintervals of P: $[0,0.2]$, $[0.2, 0.6]$. $[0.6, 1]$, $[1, 1.5]$, $[1.5, 2]$.



The lengths of the subintervals are $\Delta x_1 = 0.2$, $\Delta x_2 = 0.4$, $\Delta x_3 = 0.4$, $\Delta x_4 = 0.5$ and $\Delta x_5 = 0.5$.

The longest subinterval length is 0.5, so the norm of the partition is $||P|| = 0.5$. In this example, there are two subintervals of this length.

The Definite Integral

In this section we consider the limit of general Riemann sums as the norm of the partitions of a closed interval $[a, b]$ approaches zero. This limiting process leads us to the definition of the definite integral of a function over a closed interval $[a, b]$.

Definition on The Definite Integral

The definition of the definite integral is based on the fact that for some functions, as the norm of the partitions of $[a, b]$ approaches zero, the values of the corresponding Riemann sums approach a limiting value J.

Definition of a Definite Integral

Let f is a function defined for $a \le x \le b$, we say that a number J is the definite integral of f over $[a, b]$ and that J is the limit of the Riemann sums $\sum_{k=1}^{n} f(c_k)\Delta x_k$ if the following condition is satisfied:

Given any number $\varepsilon > 0$, there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with $||P|| < \delta$ and any choice of c_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^{n} f(c_k)\Delta x_k - J \right| < \varepsilon.$$

The definition involves a limiting process in which the norm of the partition goes to zero. When this limit exists, the function f is said to be integrable over $[a, b]$.

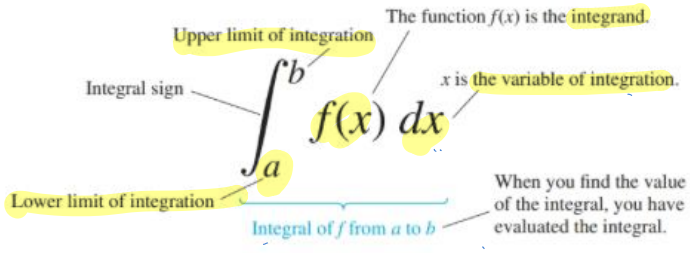
We have many choices for a partition P with norm going to zero, and many choices of points c_k for each partition. The definite integral exists when we always get the same limit J, no matter what choices are made. When the limit exists we write

$$J = \lim_{||P||\to 0} \sum_{k=1}^{n} f(c_k)\Delta x_k,$$

and we say that the *definite integral exists*.

If the definite integral exists, then instead of writing J, we write

$$\int_a^b f(x)\,dx.$$



When the definite integral exists, we say that the Riemann sums of f on $[a, b]$ **converge** to definite integral $J = \int_a^b f(x)\,dx$ and that f is **integrable** over $[a, b]$.

If equal width $\Delta x = \frac{b-a}{n}$,

$$S_n = \sum_{k=1}^{n} f(c_k)\left(\frac{b-a}{n}\right).$$

If the definite integral exists, then these Riemann sums converge to the definite integral of f over $[a, b]$, so

$$J = \int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(c_k)\left(\frac{b-a}{n}\right).$$

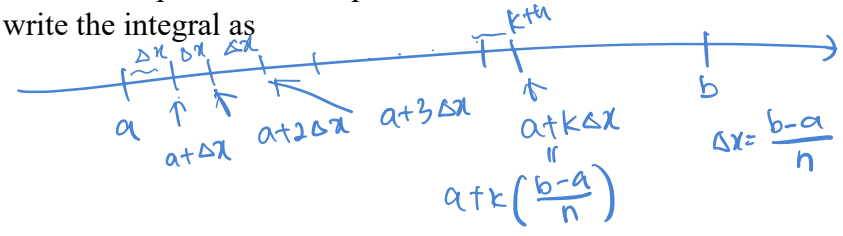
$c_k =$ _____

The Definite Integral as a Limit of Riemann Sums with Equal-Width Subintervals

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f\left(a + k\left(\frac{b-a}{n}\right)\right)\left(\frac{b-a}{n}\right).$$

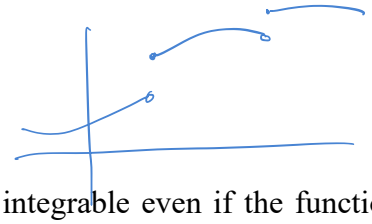
x_k (right endpoint of subinterval).

The value of the definite integral of a function over any particular interval depends on the function, not on the letter we choose to represent its independent variable. If we decide to use t or u instead of x, we simply write the integral as



$$\int_a^b f(t)\,dt \quad \text{or} \quad \int_a^b f(u)\,du \qquad \text{instead of} \qquad \int_a^b f(x)\,dx$$

No matter how we write the integral, it is still the same number, the limit of the Riemann sums as the norm of the partition approaches zero. Since it does not matter what letter we use, the variable of integration is called a **dummy variable**.



Integrable and nonintegrable Functions

Not every function defined over a closed interval $[a, b]$ is integrable even if the function is bounded. That is, the Riemann sums for some functions might not converge to the same limiting value, or to any value at all.

Theorem. If a function f is continuous over the interval $[a, b]$, or if f has at most finitely many jump discontinuities there, then the definite integral $\int_a^b f(x)\,dx$ exists and f is integrable over $[a, b]$.

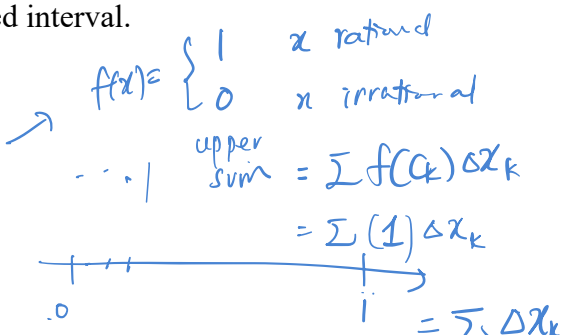
The upper and lower sums can be shown to converge to the same limiting value as the norm of the partition P tends to zero. Moreover, every Riemann sum is trapped between the values of the upper and lower sums, so every Riemann sum converges to the same limit as well. Therefore, the number J in the definition of the definite integral exists, and the continuous function f is integrable over $[a, b]$.

For integrability to fail, a function needs to be sufficiently discontinuous that the region between its graph and the x-axis cannot be approximated well by increasingly thin rectangles. Our first example is a function that is not integrable over a closed interval.

Example. The function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational,} \end{cases}$$

has no Riemann integral over [0,1].



Thus the function jumps up and down too erratically over [0,1] to allow the region beneath its graph and above the x-axis to be approximated by rectangles, no matter how thin they are.

If we choose a partition P of [0,1], then the lengths of the intervals in the partition sum to 1; $\sum_{k=1}^{n} \Delta x_k = 1$. In each subinterval $[x_{k-1}, x_k]$ there is a rational point, say c_k, $f(c_k) = 1$ (maximum value).

$$U = \sum_{k=1}^{n} f(c_k)\Delta x_k = \sum_{k=1}^{n} (1)\Delta x_k = 1.$$

Lower sum $= \sum f(c_k)\,\Delta x_k$
$= \sum 0\,\Delta x_k$
$= 0$

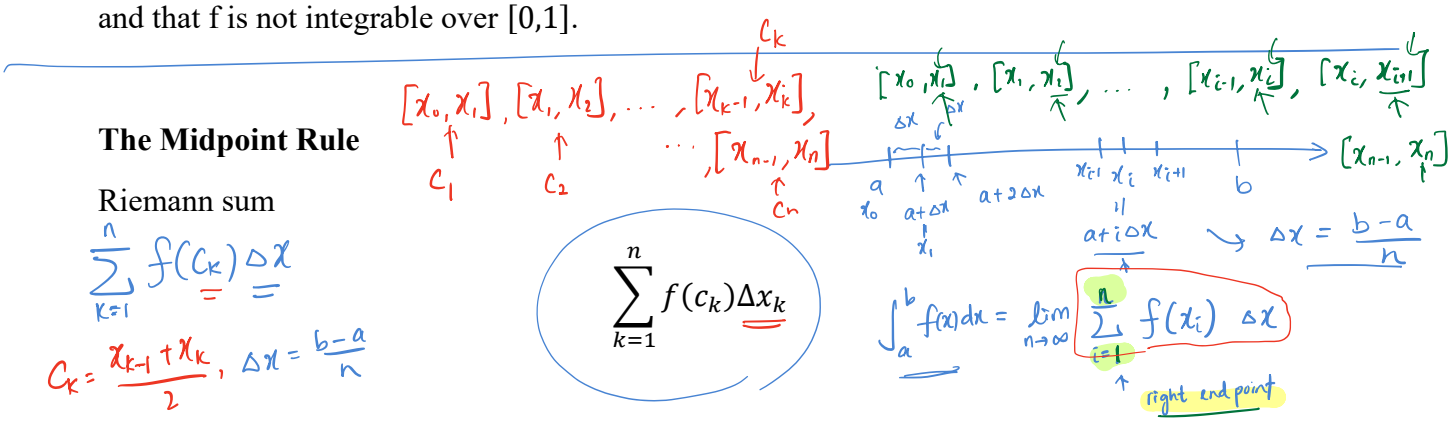
$\Rightarrow$ f is not integrable.

On the other hand, we could pick the c_k's differently to get a different result. Each subinterval $[x_{k-1}, x_k]$ also contains an irrational point c_k, $f(c_k) = 0$. The corresponding lower sum approximation is

$$L = \sum_{k=1}^{n} f(c_k)\Delta x_k = \sum_{k=1}^{n} (0)\Delta x_k = 0.$$

These lower sum approximations converge to 0 as the norm of the partition converges to 0 (because they each equal 0).

Thus making different choices for the points c_k results in different limits for the corresponding Riemann sums. We conclude that the definite integral of f over the interval $[0,1]$ does not exist and that f is not integrable over $[0,1]$.



The Midpoint Rule

Riemann sum

we have many options on how to choose a partition and where to choose a point c_k in the kth interval of the partition.

The simplest choice for a partition with n subintervals is to take each subinterval to have equal length $\Delta x = \frac{b-a}{n}$.

The midpoint is a good choice because on a small interval, the graph of a differentiable function can be approximated by a linear function, and the value of a linear function at an interval's midpoint is its average value over that interval.

With these choices, $[x_{k-1}, x_k] = [a + (k-1)\Delta x, a + k\Delta x]$ is the kth interval in the partition, its midpoint is located at $c_k = \frac{x_{k-1}+x_k}{2}$ and width is $\Delta x = \frac{b-a}{n}$.

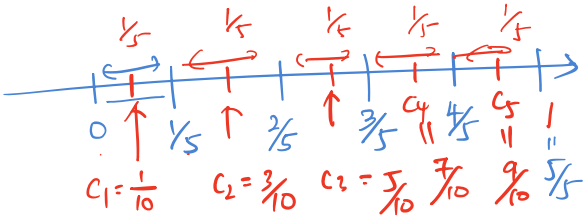
Midpoint Rule for Approximating a Definite Integral

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f(c_k)\left(\frac{b-a}{n}\right) = [f(c_1) + f(c_2) + \cdots + f(c_n)]\left(\frac{b-a}{n}\right).$$

with

$$c_k = \frac{x_{k-1} + x_k}{2} \text{ and } x_k = a + k\left(\frac{b-a}{n}\right).$$

$\int \frac{1}{1+x^2}\,dx$ = TBC.



$\frac{1-0}{5} = \frac{1}{5}$

Example. $f(x) = \frac{1}{1+x^2}$

Approximate $\int_0^1 \frac{1}{1+x^2}\,dx$ using the midpoint rule with five intervals of equal length.

$$\int_0^1 \frac{1}{1+x^2}\,dx \simeq \sum_{k=1}^{5} f(c_k)\,\Delta x \qquad \Delta x = \frac{1-0}{5} = \frac{1}{5}$$

$$\simeq \frac{1}{5}\sum_{k=1}^{5} f(c_k)$$

$$\simeq \frac{1}{5}\left[f\left(\tfrac{1}{10}\right) + f\left(\tfrac{3}{10}\right) + f\left(\tfrac{5}{10}\right) + f\left(\tfrac{7}{10}\right) + f\left(\tfrac{9}{10}\right)\right]$$

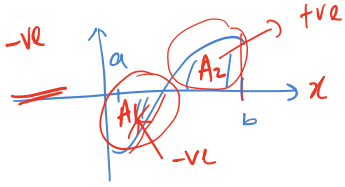
$$\simeq \frac{1}{5}\left(\frac{1}{1+\left(\frac{1}{10}\right)^2} + \frac{1}{1+\left(\frac{3}{10}\right)^2} + \frac{1}{1+\left(\frac{5}{10}\right)^2} + \frac{1}{1+\left(\frac{7}{10}\right)^2} + \frac{1}{1+\left(\frac{9}{10}\right)^2}\right)$$

$$\simeq \frac{1}{5}\left(\frac{100}{101} + \frac{100}{109} + \frac{4}{5} + \frac{100}{149} + \frac{100}{181}\right) \simeq \frac{1}{5}(3.93115733)$$

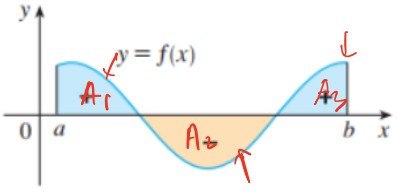
$$\simeq 0.7862.$$

Note: In the notation, $\int_a^b f(x)\,dx$, $f(x)$ is called the **integrand** and a and b are called **limits of integration**; a is the **lower limit** and b is the **upper limit**. The procedure of calculating an integral is called **integration**.

$\int_a^b f(x)\,dx$ → net area



$\int_a^b f(x)\,dx = A_1 + A_2$ (+ve) = -ve

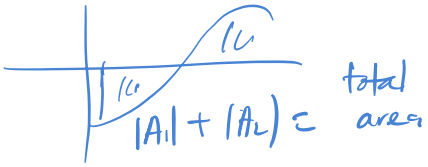


$\int_a^b f(x)\,dx$ is the net area.

$\int_a^b f(x)\,dx$: b → upper limit; f(x) dx → variable; f(x) → integrand; a → lower limit

If f takes on both positive and negative values, as in the figure above. A definite integral can be interpreted as a **net area**, that is, a difference of areas:

$$\int_a^b f(x)\,dx = A_1 - A_2$$



where A_1 is the area of the region above the x-axis and below the graph of f, and A_2 is the area of the region below the x-axis and above the graph of f. *(Note that net area is different with area under the curve)*

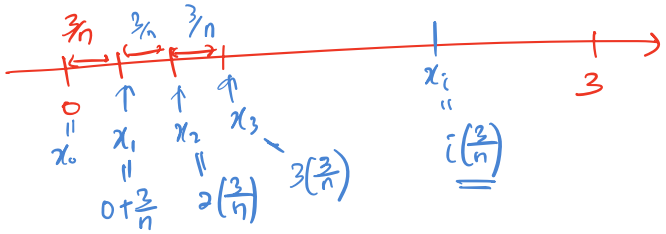
Example.

Evaluate $\int_0^3 (x^3 - 6x)\,dx$ by using right-endpoints.

$\Delta x = \frac{3-0}{n} = \frac{3}{n}$

Solution.

With n subintervals we have



$$\sum_{i=1}^{n} f(x_i)\,\Delta x$$

$$= \sum_{i=1}^{n} \left(x_i^3 - 6x_i\right)\left(\frac{3}{n}\right)$$

$$= \frac{3}{n}\sum_{i=1}^{n}\left[\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right)\right]$$

$$= \frac{3}{n}\sum_{i=1}^{n}\left(\frac{27}{n^3}(i^3) - \frac{18}{n}i\right)$$

$$= \frac{3}{n}\left(\frac{27}{n^3}\sum_{i=1}^{n} i^3 - \frac{18}{n}\sum_{i=1}^{n} i\right)$$

$$= \frac{3}{n}\left(\frac{27}{n^3}\left[\frac{n(n+1)}{2}\right]^2 - \frac{18}{n}\left[\frac{n(n+1)}{2}\right]\right)$$

$$= \frac{81}{4}\left(\frac{n^2(n+1)^2}{n^4}\right) - \frac{18}{4}(3)\left(\frac{n(n+1)}{n^2}\right)$$

$$\sum ca_k + db_k = c\sum a_k + d\sum b_k$$

$$\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\int_0^3 (x^3 - 6x)\,dx = \lim_{n\to\infty}\left[\frac{81}{4}\left(\frac{n^2(n+1)^2}{n^4}\right) - 27\left(\frac{n(n+1)}{n^2}\right)\right]$$

definite integral

$$= \lim_{n\to\infty}\left(\frac{81}{4}\left(\frac{(n+1)^2}{n^2}\right) - 27\left(\frac{n+1}{n}\right)\right)$$

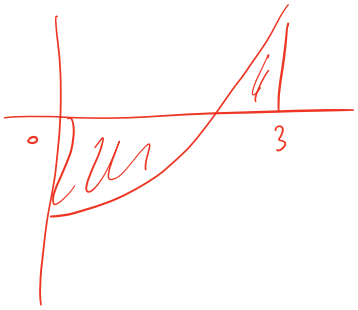
$$= \lim_{n\to\infty}\left(\frac{81}{4}\left(1+\frac{1}{n}\right)^2 - 27\left(1+\frac{1}{n}\right)\right)$$

$$= \frac{81}{4}(1)^2 - 27(1)$$

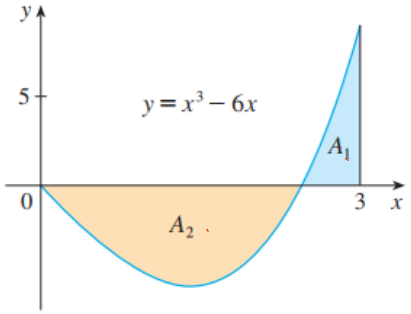
$$= -\frac{27}{4} \simeq -6.75$$

(-ve value)

$n \to \infty \quad \frac{1}{n} \to 0.$



Note: This integral can't be interpreted as an area because f takes on both positive and negative values. But it can be interpreted as the difference of areas $A_1 - A_2$, where A_1 and A_2 are shown in the following figure.



$$\int_0^3 (x^3 - 6x)\,dx = A_1 - A_2 = -6.75$$

net area ≠ total area under the curve.

Properties of the Definite Integral

When we defined the definite integral $\int_a^b f(x)\,dx$, we implicitly assumed that $a < b$. But the definition as a limit of Riemann Sum makes sense even if $a > b$. Notice that if we reverse a and b, then Δx changes from $(b-a)/n$ to $(a-b)/n$. Therefore

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

If $a = b$, then $\Delta x = 0$ and so

$$\int_a^a f(x)\,dx = 0$$

Properties of the Integral

We assume that f and g are continuous functions.

1. $\int_a^b c\,dx = c(b-a)$, where c is any constant.
2. $\int_a^b \big(f(x) \pm g(x)\big)\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$.
3. $\int_a^b k\big(f(x)\big)\,dx = k\int_a^b f(x)\,dx$, where k is any constant.

Example. Use the properties of integrals to evaluate $\int_0^1 (4+3x^2)\,dx$.

Solution. By using Properties 2 and 3. We have

$$\int_0^1 (4+3x^2)\,dx = \int_0^1 4\,dx + \int_0^1 3x^2\,dx = \int_0^1 4\,dx + 3\int_0^1 x^2\,dx.$$

From Property 1,

$$\int_0^1 4\,dx = 4(1-0) = 4$$

and

$$3\int_0^1 x^2\,dx = 3\left(\frac{1}{3}\right) = 1.$$

Hence,

$$\int_0^1 (4+3x^2)\,dx = 4+1 = 5.$$

4. $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$

Example. If it is known that $\int_0^{10} f(x)\,dx = 17$ and $\int_0^8 f(x) = 12$, find $\int_8^{10} f(x)\,dx$.

Solution. By Property 5, we have

$$\int_0^8 f(x)\,dx + \int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx.$$

So,

$$\int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx - \int_0^8 f(x)\,dx = 17 - 12 = 5.$$

Comparison Properties of the Integral

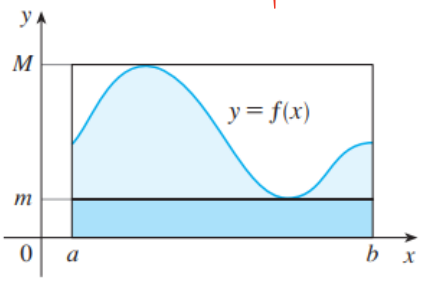
6. *Max-Min Inequality*: If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then

$$(\min f)\,(b-a) \le \int_a^b f(x)\,dx \le (\max f)\,(b-a).$$

7. *Domination*: if $f(x) \ge g(x)$ on $[a, b]$, then $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$.

If $f(x) \ge 0$ on $[a, b]$, then $\int_a^b f(x)\,dx \ge 0$.

Proof of Property 6.



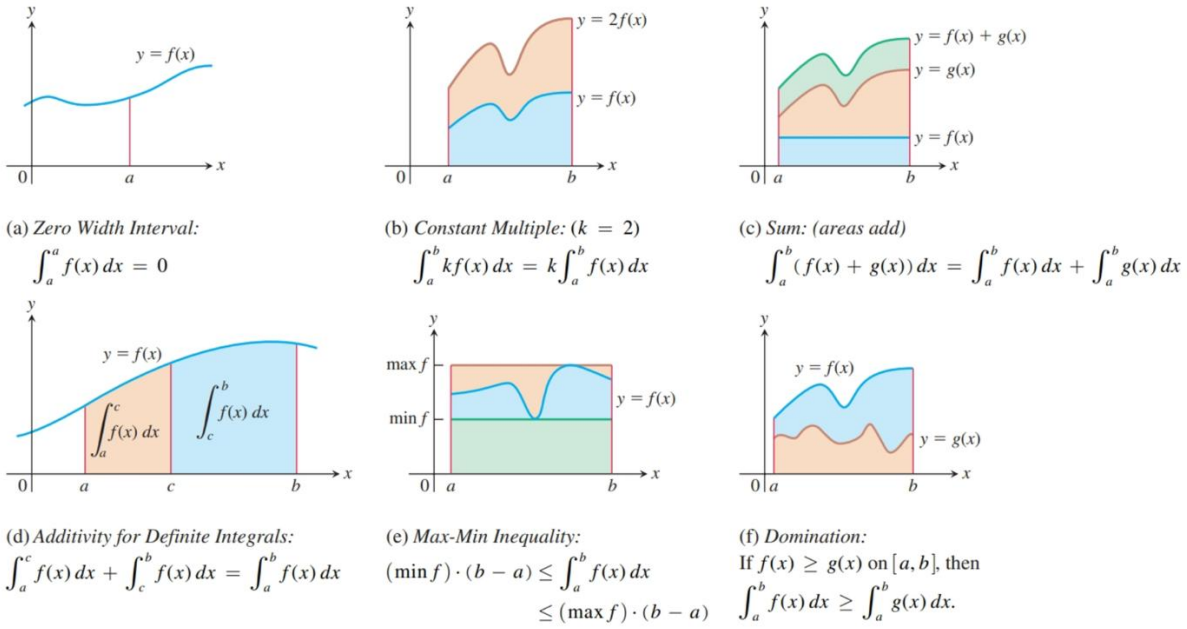
Since $m \le f(x) \le M$, Property 7 gives

$$\int_a^b m\,dx \le \int_a^b f(x)\,dx \le \int_a^b M\,dx$$

Using Property 1 to evaluate the integrals on the left and right sides, we obtain

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

Property 8 is useful when we want a rough estimate of the size of an integral.



(a) *Zero Width Interval:* $\int_a^a f(x)\,dx = 0$

(b) *Constant Multiple:* $(k = 2)$ $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$

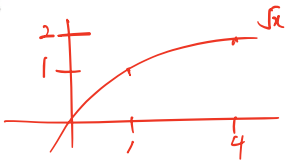
(c) *Sum: (areas add)* $\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

(d) *Additivity for Definite Integrals:* $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$

(e) *Max-Min Inequality:* $(\min f)\cdot(b-a) \le \int_a^b f(x)\,dx \le (\max f)\cdot(b-a)$

(f) *Domination:* If $f(x) \ge g(x)$ on $[a,b]$, then $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$.

FIGURE 5.11 Geometric interpretations of Rules 2–7 in Table 5.6.



$1 \le \sqrt{x} \le 2$, $[1,4]$

$1(4-1) \le \int_1^4 \sqrt{x}\,dx \le 2(4-1)$

$3 \le \int_1^4 \sqrt{x}\,dx \le 6$

Example. Use Property 8 to estimate $\int_1^4 \sqrt{x}\,dx$.

Solution. Since $f(x) = \sqrt{x}$ is an increasing function, its absolute minimum on $[1,4]$ is $m = f(1) = 1$ and its absolute maximum on $[1,4]$ is $M = f(4) = 2$. Thus,

$$1(4-1) \le \int_1^4 \sqrt{x}\,dx \le 2(4-1)$$

$$3 \le \int_1^4 \sqrt{x}\,dx \le 6.$$

$\int_1^4 \sqrt{x}\,dx = \frac{2}{3}\left[x^{3/2}\right]_1^4$

$= \frac{2}{3}(2^3 - 1)$

$= \frac{2}{3}(7)$

≈ 4.6667

$4\frac{2}{3}$

Example. Show that the value of $\int_0^1 \sqrt{1+\cos x}\,dx$ is less than or equal to $\sqrt{2}$.

$\cos x \le 1$

$1 + \cos x \le 2$

$\sqrt{1+\cos x} \le \sqrt{2}$

$\int_0^1 \sqrt{1+\cos x}\,dx \le \int_0^1 \sqrt{2}\,dx = \sqrt{2}$.

$f(x) = \sqrt{1+\cos x}$

$f(x) \le \sqrt{2}$.

Area Under the Graph of a Nonnegative Function

We now return to the problem that started this chapter, which is defining what we mean by the area of a region having a curved boundary.

We approximated the area under the graph of a nonnegative continuous function using several types of finite sums of areas of rectangles that approximate the region—upper sums, lower sums, and sums using the midpoints of each subinterval—all of which are Riemann sums constructed in special ways.

As a result, we can now define the area under the graph of a nonnegative integrable function to be the value of that definite integral.

Definition

If $y = f(x)$ is nonnegative and integrable over a closed interval $[a, b]$, then the area under the curve $y = f(x)$ over $[a, b]$ is the integral of f from a to b,

$$A = \int_a^b f(x)\,dx.$$

For the first time, we have a rigorous definition for the area of a region whose boundary is the graph of a continuous function. We now apply this to a simple example, the area under a straight line, and we verify that our new definition agrees with our previous notion of area.

Example. Compute $\int_0^b x\,dx$ and find the area A under $y = x$ over the interval $[0, b]$, where $b > 0$.

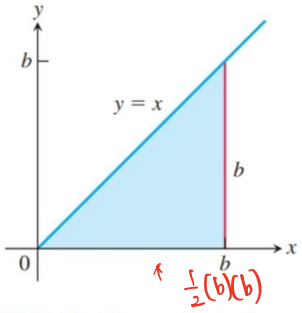
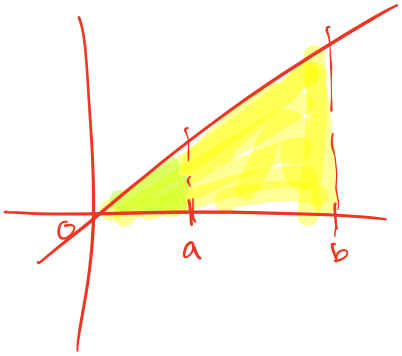


FIGURE 5.12 The region in Example 5 is a triangle.

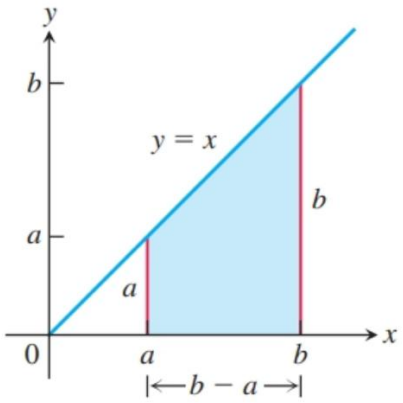
Example can be generalized to integrate $f(x) = x$ over any closed interval $[a, b]$ for which $0 < a < b$.



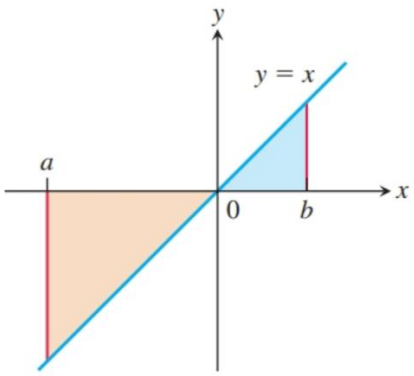
In conclusion, we have the following rule for integrating $f(x) = x$ over the interval $[a, b]$:

$$\int_a^b x\,dx = \frac{b^2}{2} - \frac{a^2}{2}, \quad a < b$$

This computation gives the area of the trapezoid in Figure 5.13a. Equation (2) remains valid when a and b are negative, but the interpretation of the definite integral changes.



(a)



(c)

FIGURE 5.13 (a) The area of this trapezoidal region is $A = (b^2 - a^2)/2$. (b) The definite integral in Equation (2) gives the negative of the area of this trapezoidal region. (c) The definite integral in Equation (2) gives the area of the blue triangular region added to the negative of the area of the tan triangular region.

When $a < 0$ and $b > 0$, Equation (2) is still valid and the definite integral gives the difference between two areas, the area under the graph and above $[b, 0]$, minus the area below $[a, 0]$ and over the graph.

The following results can also be established by using a Riemann sum calculation similar to the one we use just now.

$$\int_a^b c\,dx = c(b - a), c \text{ constant}$$

$$\int_a^b x^2 dx = \frac{b^3}{3} - \frac{a^3}{3}, a < b.$$

$\int_a^b x^2\,dx = \frac{b^3}{3} - \frac{a^3}{3}$.

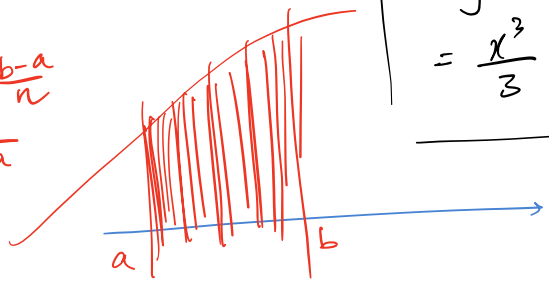
Antiderivative

$\int x^2\,dx = \frac{x^3}{3} + C$

Average $\begin{cases} f(c_1) : \\ f(c_2) = \\ \vdots \\ n \text{ term} \end{cases}$

$\Delta x = \frac{b-a}{n}$

$\frac{1}{n} = \frac{\Delta x}{b-a}$



Average Value of a Continuous Function Revisited

We informally introduced the average value of a nonnegative continuous function f over an interval $[a, b]$, leading us to define this average as the area under the graph of $y = f(x)$ divided by $b - a$.

$$\text{Average} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

This formula gives us a precise definition of the average value of a continuous (or integrable) function, whether it is positive, negative, or both.

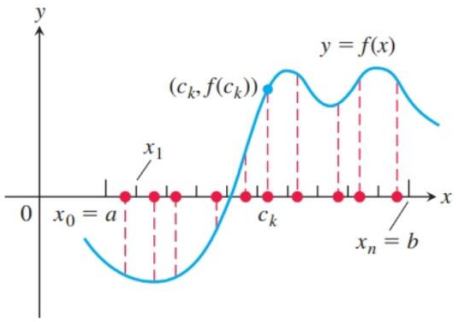


FIGURE 5.14 A sample of values of a function on an interval $[a, b]$.

We start with the idea from arithmetic that the average of n numbers is their sum divided by n. A continuous function f on $[a, b]$ may have infinitely many values.

$$\frac{f(c_1) + f(c_2) + \cdots + f(c_n)}{n} = \frac{1}{n}\sum_{k=1}^{n} f(c_k)$$
$$= \frac{\Delta x}{b-a}\sum_{k=1}^{n} f(c_k)$$
$$= \frac{1}{b-a}\sum_{k=1}^{n} f(c_k)\ \Delta x$$

The average of the samples is obtained by dividing a Riemann sum for f on $[a, b]$ by $(b - a)$. As we increase the number of samples and let the norm of the partition approach to zero, the average approaches $\left(\frac{1}{b-a}\right)\int_a^b f(x)\,dx$.

Definition.

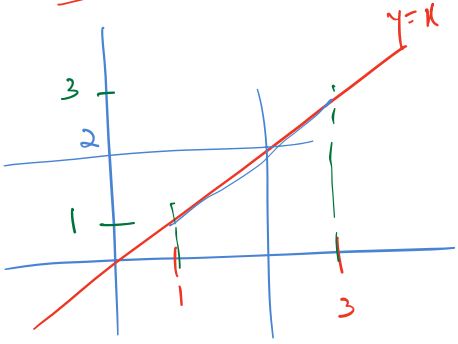
If f is integrable on $[a, b]$, then its **average value** on $[a, b]$, also called its **mean**, is

$$\text{av}(f) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

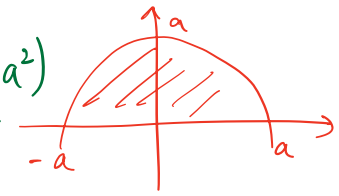
Example. Find the average value of $f(x) = x$ on $[1,3]$.

$$\text{Av}(f) = \frac{1}{3-1}\int_1^3 x\,dx$$
$$= \frac{1}{2}\left(\frac{3^2}{2} - \frac{1^2}{2}\right)$$
$$= \frac{1}{2}\left(\frac{9}{2} - \frac{1}{2}\right)$$
$$= \frac{1}{2}\left(\frac{8}{2}\right) = 2.$$

$$\int_a^b x\,dx = \frac{b^2}{2} - \frac{a^2}{2}.$$



$$\int_{-a}^{a} \sqrt{a^2-x^2}\,dx = \frac{1}{2}(\pi a^2) = \frac{\pi a^2}{2}$$



Example. Find the average value of $f(x) = \sqrt{4-x^2}$ on $[-2,2]$.

$0 \le f(x) = \sqrt{4-x^2}$, $[-2,2]$

$$\int_{-2}^{2} \sqrt{4-x^2}\,dx = \frac{1}{2}(\pi(2)^2)$$
$$= 2\pi$$

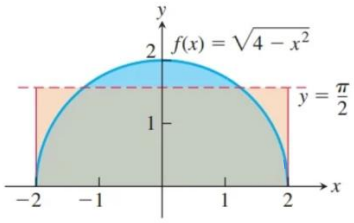


FIGURE 5.15 The average value of $f(x) = \sqrt{4-x^2}$ on $[-2,2]$ is $\pi/2$ (Example 7). The area of the rectangle shown here is $4 \cdot (\pi/2) = 2\pi$, which is also the area of the semicircle.

$$Av(f) = \frac{1}{2-(-2)} \int_{-2}^{2} \sqrt{4-x^2}\,dx$$
$$= \frac{1}{4}(2\pi)$$
$$= \frac{\pi}{2}.$$

integration ↕ differentiation
derivation

Fundamental Theorem of Calculus

$F(x)$ antiderivative of $f(x)$ $\quad y \rightleftarrows \frac{dy}{dx}$

$\frac{d}{dx}F(x) = f(x)$ $\quad$ indefinite integral → Antiderivation $\quad \int f(x)\,dx = F(x)+C.$

In this section, we present the Fundamental Theorem of Calculus, which is the central theorem of integral calculus. It connects integration and differentiation, enabling us to compute integrals by using an antiderivative of the integrand function, rather than by taking limits of Riemann sums as we did.

Mean Value Theorem for Definite Integrals

MVT

f continuous $[a,b]$

f differentiable (a,b)

Then, $\exists c \in (a,b)$ s.t.

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

↑ average "mean"

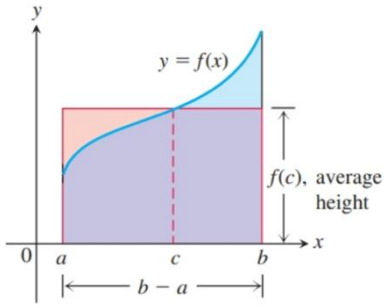
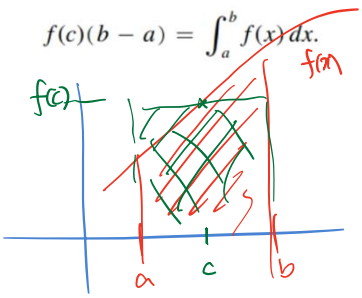


FIGURE 5.16 The value $f(c)$ in the Mean Value Theorem is, in a sense, the average (or *mean*) height of f on $[a,b]$. When $f \ge 0$, the area of the rectangle is the area under the graph of f from a to b,

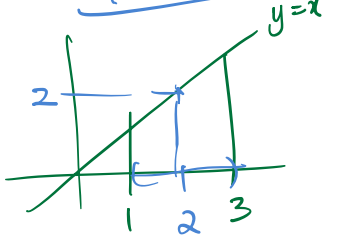
$$f(c)(b-a) = \int_a^b f(x)\,dx.$$



Geometrically, the Mean Value Theorem says that there is a number c in $[a,b]$ such that the rectangle with height equal to the average value $f(c)$ of the function and base width $b-a$ has exactly the same area as the region beneath the graph of f from a to b.

$$f(c)(b-a) = \int_a^b f(x)\,dx$$

$\exists c \in (a,b)$



$$\int_1^3 x\,dx = \frac{3^2}{2} - \frac{1}{2}$$
$$= 8/2 = 4$$

$$f(c)(3-1) = 4$$
$$2(2) = 4$$

$\exists c \longrightarrow f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$

$\underbrace{\qquad\qquad}_{av(f)}$

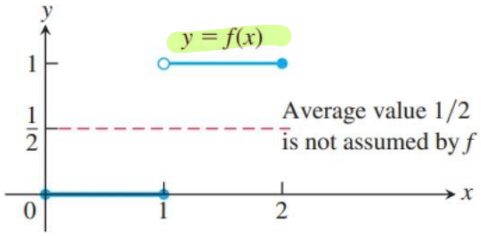


FIGURE 5.17 A discontinuous function need not assume its average value.

Theorem The Mean Value Theorem for Definite Integrals

If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Example. Show that if f is continuous on $[a, b]$, $a \neq b$ and if

$$\int_a^b f(x)\,dx = 0,$$

Then $f(x) = 0$ at least once in $[a, b]$.

f continuous $[a,b]$

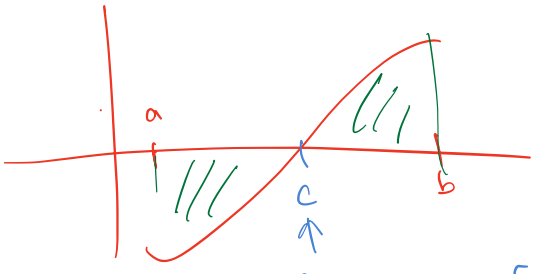
By Mean Value Theorem of integration,

$\exists c \in (a,b)$ s.t

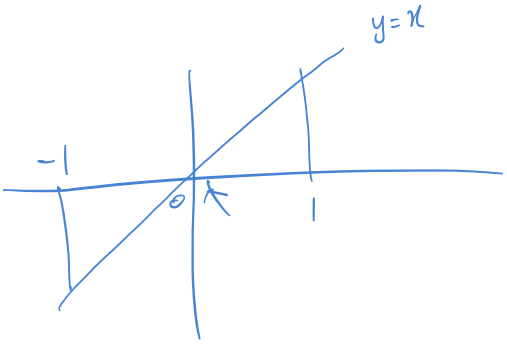
$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$

$= \frac{1}{b-a}(0)$

$= 0.$



$f(c) = 0$, $c \in (a,b)$



$f(x) = x$

$\int_{-1}^{1} f(x)\,dx$

$= \int_{-1}^{1} x\,dx$

$= \frac{x^2}{2}\Big|_{-1}^{1}$

$= \frac{1}{2} - \frac{1}{2} = 0$

Mean Value Theorem for integration

$\exists c \in (-1, 1)$ s.t.

$f(c) = 0$

$$\int_0^2 9x^2-16x+4\,dx$$
$$=\left[\frac{9x^3}{3}-\frac{16x^2}{2}+4x\right]_0^2$$
$$=3(8)-8(4)+4(2)$$
$$=0$$

Mean value Theorem,
$\exists c \in (0,2)$ s.t. $f(c)=0$

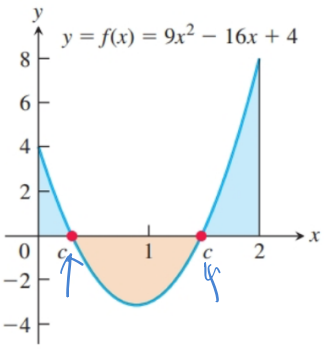


FIGURE 5.18 The function $f(x) = 9x^2 - 16x + 4$ satisfies $\int_0^2 f(x)\,dx = 0$, and there are two values of c in the interval $[0, 2]$ where $f(c) = 0$.

This is illustrated in Figure 5.18 for the function $f(x) = 9x^2 - 16x + 4$ on the interval $[0,2]$.

Fundamental Theorem, Part I

It can be very difficult to compute definite integrals by taking the limit of Riemann sums. We now develop a powerful new method for evaluating definite integrals, based on using antiderivatives.

If $f(t)$ is an integrable function over a finite interval I, then the integral from any fixed number $a \in I$ to another number $x \in I$ defines a new function F whose value at x is

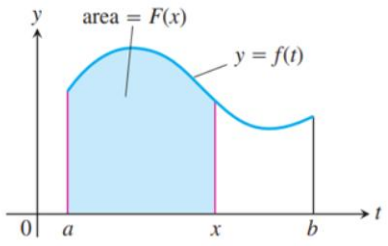
$$F(x) = \int_a^x f(t)\,dt.$$

Equation above gives a way to define new functions, but its importance now is the connection it makes between integrals and derivatives.

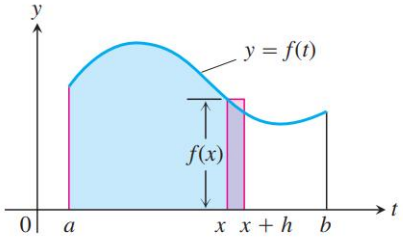
If f is any continuous function, then the Fundamental Theorem asserts that F is a differentiable function of x whose derivative is f itself. At every value of x,

$$\frac{d}{dx}F(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

We look at the geometry behind it.



The function $F(x)$ gives the area under the graph of f from a to x when f is nonnegative and $x > a$.



$F(x)$ is the area to the left of x. Also, $F(x + h)$ is the area to the left of $x + h$. The difference quotient $[F(x + h) - F(x)]/h$ is then approximately equal to $f(x)$, the height of the rectangle shown here.

If $f \geq 0$ on $[a.b]$, then the computation of $F'(x)$ from the definition of the derivative means taking the limit as $h \to 0$ of the difference quotient

$$\frac{F(x+h) - F(x)}{h}.$$

For $h > 0$, the numerator is obtained by subtracting two areas, so it is the area under the graph of f from x to $x + h$. If h is small this area is approximately equal to the area of the rectangle of height $f(x)$ and width h (see the figure above).

That is, $F(x + h) - F(x) \approx hf(x)$. Dividing both sides of this approximation by h and letting $h \to 0$, it is reasonable to expect that

$$F'(x) = \lim_{h \to 0} \frac{F(x+h) - F(x)}{h} = f(x).$$

This result is true even if the function f is not positive, and this forms the first part of the Fundamental Theorem of Calculus.

$F(x) = \int_a^x f(t)\,dt \Rightarrow \frac{d}{dx}F(x) = f(x)$

$F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x)$

Theorem. *The Fundamental Theorem of Calculus, Part 1*

If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)\,dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$;

differentiation = anti-integration

$\frac{d}{dx}F(x) = f(x)$

$$F'(x) = \frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x).$$

Example. *Applying the Fundamental Theorem*

Use Fundamental Theorem to find $\frac{dy}{dx}$ if

(a) $y = \int_a^x \cos t\,dt$

(b) $y = \int_a^x \frac{1}{1+t^2}\,dt$

(c) $y = \int_x^5 3t \sin t\,dt$

(d) $y = \int_1^{x^2} \cos t\,dt$

(e) $y = \int_a^x (t^3 + 1)\,dt$

(f) $y = \int_{1+3x^2}^4 \frac{1}{2+t}\,dt$

Solution.

(a) $\frac{d}{dx}\int_a^x \cos t\,dt = \cos x$

(b) $\frac{d}{dx}\int_a^x \frac{1}{1+t^2}\,dt = \frac{1}{1+x^2}$

$\int_a^x x\,u\,du = x\int_a^x u\,du$ (x is constant)

$F(x) = \int_a^x f(t)\,dt$

(c) $y = \int_x^5 3t\sin t\,dt = -\int_5^x 3t\sin t\,dt$

$\frac{dy}{dx} = -\frac{d}{dx}\int_5^x 3t\sin t\,dt$

$= -3x\sin x$

(d) $y = \int_1^{x^2} \cos t\,dt$

$u = x^2$

$y = \int_1^u \cos t\,dt$

$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$

$= \frac{d}{du}\int_1^u \cos t\,dt \cdot \frac{du}{dx}$

$= \cos u \cdot (2x)$

$= 2x\cos(x^2)$

(e) $y = \int_a^x t^3 + 1\,dt$

$\frac{dy}{dx} = x^3 + 1$

(f) $y = \int_{1+3x^2}^4 \frac{1}{2+t}\,dt$

$= -\int_4^{1+3x^2} \frac{1}{2+t}\,dt = -\int_4^u \frac{1}{2+t}\,dt$

$u = 1+3x^2 \quad \frac{du}{dx} = 6x$

$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx} = \frac{-1}{2+u}(6x) = \frac{-6x}{2+(1+3x^2)}$.

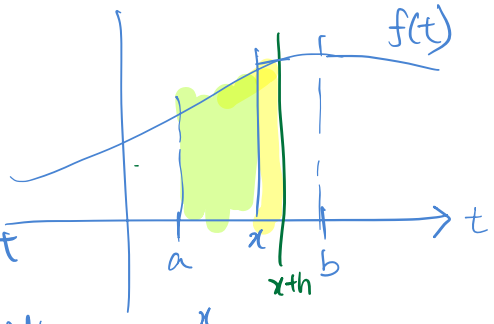
$\int_1^x t\,dt = \frac{t^2}{2}\Big|_1^x$

$F(x) = \frac{x^2}{2} - \frac{1}{2}$

$F(x) = \int_a^x f(t)\,dt$

$F(x+h) = \int_a^{x+h} f(t)\,dt$

$F(x) = \int_a^x f(t)\,dt$



$F(x^2) = \int_1^{x^2} \cos t\,dt$

$\frac{d}{dx}F(x) = \lim_{h\to 0}\left(\frac{F(x+h)-F(x)}{h}\right)$

$= \lim_{h\to 0}\frac{\int_x^{x+h} f(t)\,dt}{h}$

$= \lim_{h\to 0}\frac{h(f(x+h))}{h}$

$= f(x)$.

$y = F(u),\ u = x^2 \quad F(x^2)$

$y = \int_1^{x^2} \cos t\,dt$

$\frac{dy}{dx} = \cos(x^2)\cdot(2x)$

$= 2x\cos(x^2)$.

(c) $\frac{dy}{dx} = \frac{d}{dx}\int_x^5 3t\sin t\,dt = \frac{d}{dx}\left(-\int_5^x 3t\sin t\,dt\right) = -3x\sin x$

(d) $y = \int_1^{x^2} \cos t\,dt$, the upper limit of integration is not x but x^2. This make y a composite of two functions, $y = \int_1^u \cos t\,dt$ and $u = x^2$. We apply Chain Rule when finding $\frac{dy}{dx}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du}\cdot\frac{du}{dx}\\ &= \left(\frac{d}{du}\int_1^u \cos t\,dt\right)\cdot\frac{du}{dx}\\ &= \cos u\cdot\frac{du}{dx}\\ &= \cos(x^2)\cdot(2x)\\ &= 2x\cos x^2.\end{aligned}$$

Fundamental Theorem, Part 2 (The Evaluation Theorem)

We now come to the second part of the Fundamental Theorem of Calculus. This part describes how to evaluate definite integrals without having to calculate limits of Riemann sums. Instead we find and evaluate an antiderivative at the upper and lower limits of integration.

Theorem. *The Fundamental Theorem of Calculus Part 2*

If f is continuous at every point of $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

The theorem says that to calculate the definite integral of f over $[a, b]$ all we need to do is:

1. Find an antiderivative F of f, and

2. Calculate the number $\int_a^b f(x)\,dx = F(b) - F(a)$.

The usual notation for $F(b) - F(a)$ is $F(x)]_a^b$ or $[F(x)]_a^b$, depending on whether F has one or more terms.

Example. *Evaluating Integrals*

(a) $\int_0^{\pi} \cos x\,dx = \sin x\,]_0^{\pi} = \sin\pi - \sin 0 = 0$

(b) $\int_1^4 \left(\frac{3}{2}\sqrt{x} - \frac{4}{x^2}\right)dx = \left[x^{\frac{3}{2}} + \frac{4}{x}\right]_1^4 = \left[(4)^{\frac{3}{2}} + \frac{4}{4}\right] - \left[1^{\frac{3}{2}} - \frac{4}{1}\right] = [8+1] - [5] = 4.$

(c) $\int_{-\frac{\pi}{4}}^0 \sec x\tan x\,dx = \sec x\,|_{-\frac{\pi}{4}}^0 = \sec 0 - \sec\left(\frac{-\pi}{4}\right) = 1 - \sqrt{2}.$

The process used in Example was much easier than a Riemann Sum computation.

The conclusions of the Fundamental Theorem tell us several things. First,

$$\frac{d}{dx}\int_a^x f(t)\,dt = \frac{dF}{dx} = f(x),$$

which mean **if you first integrate the function f then differentiate the result, you get the function f back again**. Likewise, the equation

$$\int_a^x \frac{dF}{dt}\,dt = \int_a^x f(t)\,dt = F(x) - F(a),$$

says that **if you first differentiate the function F and then integral the result, you get the function F back** (adjusted by an integration constant).

The processes of integration and differentiation are "inverses" of each other. The Fundamental Theorem also says that every continuous function f has an antiderivative F. And it says that the differential equation $\frac{dy}{dx} = f(x)$ has a solution (namely, the function $y = F(x)$) for every continuous function f.

Indefinite Integrals and the Substitution Rule

The set of all antiderivatives of the function f is called the **indefinite integral** of f with respect to x, and is symbolized by

$$\int f(x)\,dx.$$

A definite integral $\int_a^b f(x)\,dx$ is a *number*. An indefinite integral $\int f(x)dx$ is a *function* plus an arbitrary constant C. When finding the indefinite integral of a function f, remember that it always includes an arbitrary constant C.

$u=f(x)$ $\int x^n dx = \frac{x^{n+1}}{n+1}+C$ $\frac{d}{dx}x^{n+1} = (n+1)x^n$ $\int u^n du$ $\Rightarrow \frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$

The Power Rule in Integral Form

If u is any differentiable function, then

$$\int u^n\,du = \frac{u^{n+1}}{n+1} + C, \qquad (n \neq -1, n \text{ rational}).$$

$\int u^n du = \frac{u^{n+1}}{n+1}+C$ $u=f(x)$

Example. *Using the Power Rule*

$\int \sqrt{1+y^2}\cdot 2y\,dy$

Solution. Let $u = 1 + y^2$. Then $\frac{du}{dy} = 2y$.

$$\int \sqrt{1+y^2}\cdot 2y\,dy = \int \sqrt{u}\,du = \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$$

$\int x\sqrt{x^2+1}\,dx$ $= \int \sqrt{x^2+1}\,(x\,dx)$ $= \int \sqrt{u}\,\frac{1}{2}du$ $= \frac{1}{2}\int \sqrt{u}\,du$ $= \frac{1}{2}\int u^{1/2}du$ $\rightsquigarrow \int x^{1/2}dx$ $= \frac{1}{2}\left(\frac{u^{3/2}}{3/2}\right)+C$ $= \frac{1}{3}(x^2+1)^{3/2}+C$.

$u=x^2+1$ $\frac{du}{dx} = 2x$ $\int \frac{du}{dx}\cdot dx = \int 2x\,dx$. $u=x^2+1$ $du = 2x\,dx$ $\frac{1}{2}du = x\,dx$

$\int \frac{3x}{(2x^2+1)^{3/2}}dx$ $u^2 = 2x^2+1$ $2u\,du = 4x\,dx$ $u\,du = 2x\,dx$

$\int \frac{3x}{(2x^2+1)^{3/2}}dx$ $= 3\int \frac{x\,dx}{(2x^2+1)^{3/2}}$ $u=2x^2+1$ $du = 4x\,dx$

$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$

$\int f(x)(f(x))^n\,dx = \frac{(f(x))^{n+1}}{n+1} + C$

$\int \frac{3x}{(2x^2+1)^{3/2}}\,dx$

$= \int 3x\,(2x^2+1)^{-3/2}\,dx$

$= \frac{3}{4}\int 4x\,(2x^2+1)^{-3/2}\,dx$

$= \frac{3}{4}\frac{(2x^2+1)^{-1/2}}{-1/2} + C = -\frac{3}{2}(2x^2+1)^{-1/2} + C$

$= \frac{3}{2}\int \frac{2x\,dx}{[(2x^2+1)^{1/2}]^3}$

$= \frac{3}{2}\int \frac{u\,du}{u^3}$

$= \frac{3}{2}\int \frac{1}{u^2}\,du$

$= \frac{3}{2}\left(\frac{u^{-1}}{-1}\right) + C$

$= -\frac{3}{2}\left(\frac{1}{(2x^2+1)^{1/2}}\right) + C.$

$= \frac{3}{4}\int \frac{4x\,dx}{(2x^2+1)^{3/2}}$

$= \frac{3}{4}\int \frac{du}{u^{3/2}} = \frac{3}{4}\int u^{-3/2}\,du$

$= \frac{3}{4}\frac{u^{-1/2}}{-1/2} + C = -\frac{3}{2}\frac{1}{\sqrt{u}} + C$

$= -\frac{3}{2}\left(\frac{1}{\sqrt{2x^2+1}}\right) + C.$

Example. *Adjusting the Integrand by a Constant*

$\int \sqrt{4t-1}\,dt$

Solution. Let $u = 4t - 1$. Then $\frac{du}{dt} = 4$.

$$\int \sqrt{4t-1}\,dt = \frac{1}{4}\int 4\sqrt{4t-1}\,dt$$
$$= \frac{1}{4}\int \sqrt{u}\,du$$
$$= \frac{1}{4}\cdot\frac{2}{3}u^{\frac{3}{2}} + C$$
$$= \frac{1}{6}(4t-1)^{\frac{3}{2}} + C$$

$\int \sqrt{4t-1}\,dt$

$= \frac{1}{4}\int 4\sqrt{4t-1}\,dt$

$= \frac{1}{4}\left(\frac{(4t-1)^{3/2}}{3/2}\right) + C$

$= \frac{1}{3}(4t-1)^{3/2} + C$

$\int \sqrt{4t-1}\,dt$

$u^2 = 4t-1$

$2u\,du = 4\,dt$

$u\,du = 2\,dt$

$\int u\left(\frac{u}{2}\,du\right)$

$= \frac{1}{2}\int u^2\,du$

$= \frac{1}{2}\left[\frac{u^3}{3}\right] + C$

$= \frac{1}{6}(4t-1)^{3/2} + C.$

$\int_1^4 \sqrt{4t-1}\,dt$ $\quad (2u = 4\,dt)$

$4t-1 \geq 0$ $\quad u^2 = 4t-1$

$t=1,\ u^2 = 3 \Rightarrow u = \pm\sqrt{3}$

$t=4,\ u^2 = 15 \Rightarrow u = \pm\sqrt{15}$

$= \int_{\sqrt{3}}^{\sqrt{15}} u\left(\frac{u}{2}\right)du$

$= \frac{1}{2}\int_{\sqrt{3}}^{\sqrt{15}} u^2\,du = \frac{1}{2}\left[\frac{u^3}{3}\right]_{\sqrt{3}}^{\sqrt{15}}$

$= \frac{1}{2}\left(\frac{15\sqrt{15}}{3} - \frac{3\sqrt{3}}{3}\right)$

$= ____.$

Example. Find the integral $\int (x^3+x)^5(3x^2+1)\,dx$.

$\int (x^3+x)^5(3x^2+1)\,dx$

$= \frac{(x^3+x)^6}{6} + C.$

$u = x^3 + x$

$du = 3x^2 + 1\,dx$

$\int u^5\,du$

$= \frac{u^6}{6} + C$

$= \frac{1}{6}(3x^2+1)^6 + C.$

$\int f'(x)(f(x))^n\,dx = \frac{(f(x))^{n+1}}{n+1} + C$

Example. Find $\int \sqrt{2x+1}\,dx$.

$\int \sqrt{2x+1}\,dx$

$= \frac{1}{2}\int 2\sqrt{2x+1}\,dx$

$= \frac{1}{2}\frac{(2x+1)^{3/2}}{3/2} + C$

$= \frac{1}{3}\sqrt{2x+1} + C$

$\int \sqrt{2x+1}\,dx$

$u = 2x+1$

$du = 2\,dx$

$\frac{1}{2}\int \sqrt{2x+1}\,(2\,dx).$

$= \frac{1}{2}\int \sqrt{u}\,du$

$= \frac{1}{2}\left(\frac{u^{3/2}}{3/2}\right) + C$

$= \frac{1}{3}u^{3/2} + C$

$= \frac{1}{3}(2x+1)^{3/2} + C.$

$\int f(g(x)) \cdot g'(x)\,dx$ $\quad u = g(x) \Rightarrow du = g'(x)\,dx$

$\int f(u)\,du$

Substitution: Running the Chain Rule Backwards

Theorem. *The Substitution Rule*

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

The *Substitution Rule* provides the following method to evaluate the integral

$$\int f(g(x))g'(x)\,dx.$$

when f and g' are continuous functions:

1. Substitute $u = g(x)$ and $du = \left(\frac{du}{dx}\right)dx = g'(x)dx$ to obtain the integral $\int f(u)\,du$.

2. Integrate with respect to u.

3. Replace u with $g(x)$ in the result.

Example. Find the integral $\int \sec^2(5x+1) \cdot 5\,dx$.

$\int \sec^2(5x+1) \cdot 5\,dx$

$f(5x+1)$ $\quad \frac{d}{dx}(5x+1)$

$f(g(x))$ $\quad g'(x)$

$u = 5x+1$

$du = 5\,dx$

$= \int \sec^2 u\,du$

$= \tan u + C$

$= \tan(5x+1) + C$

Example. *Using Substitution*

$$\int \cos(7\theta+5)\,d\theta = \int \cos u \left(\frac{1}{7}du\right)$$

$$= \frac{1}{7}\int \cos u\,du$$

$$= \frac{1}{7}(\sin u) + C$$

$$= \frac{1}{7}\sin(7\theta+5) + C$$

$f(g(x))$

$\int \cos(7\theta+5)\,d\theta$

$g(x) = 7\theta+5$ $\quad g'(x) = 7$

$= \frac{1}{7}\int \cos(7\theta+5) \cdot 7\,d\theta$

$f(g(x))$ $\quad g'(x)$

$= \frac{1}{7}\int \cos u\,du$

$= \frac{1}{7}(\sin u) + C$

$= \frac{1}{7}\sin(7\theta+5) + C$

Example. Find the integral $\int x^2 \cos x^3 \, dx$.

$u = x^3 \Rightarrow du = 3x^2 dx$

$\int x^2 \cos(x^3) dx = \frac{1}{3}\int \cos(x^3)(3x^2 dx)$

$= \frac{1}{3}\int \cos u \, du$

$= \frac{1}{3}(\sin u) + C = \frac{1}{3}\sin(x^3) + C$.

Example. Evaluate $\int x\sqrt{2x+1}\, dx$.

$\int x\sqrt{2x+1}\, dx$

$u^2 = 2x+1$

$2u\, du = 2\, dx$

$u\, du = dx$

$\int \left(\frac{u^2-1}{2}\right) u \,(u\, du)$

$= \frac{1}{2}\int u^4 - u^2 \, du$

$= \frac{1}{2}\left[\frac{u^5}{5} - \frac{u^3}{3}\right] + C$

$= \frac{1}{2}\left(\frac{1}{5}(2x+1)^{5/2} - \frac{1}{3}(2x+1)^{3/2}\right) + C$

$= \frac{1}{2}\int (2x+1)\sqrt{2x+1}\, dx - \frac{1}{2}\int \sqrt{2x+1}\, dx$

$= \frac{1}{2}\int (2x+1)^{3/2} dx - \frac{1}{2}\int (2x+1)^{1/2} dx$

$= \frac{1}{4}\int 2(2x+1)^{3/2} dx - \frac{1}{4}\int 2(2x+1)^{1/2} dx$

$= \frac{1}{4}\frac{(2x+1)^{5/2}}{5/2} - \frac{1}{4}\frac{(2x+1)^{3/2}}{3/2} + C$

$= \frac{1}{10}(2x+1)^{5/2} - \frac{1}{6}(2x+1)^{3/2} + C$.

The Integrals of $\sin^2 x$ and $\cos^2 x$.

Sometimes we can use trigonometric identities to transform integrals we do not know how to evaluate into ones we can using the substitution rule.

$\int \sin^2 x \, dx$

$\int \cos^2 x \, dx$

$\cos^2 x + \sin^2 x = 1$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\int \cos Ax \, dx = \frac{\sin Ax}{A} + C$

$\int \sin Ax \, dx = -\frac{\cos Ax}{A} + C$

Example.

(a) $\int \sin^2 x \, dx = \int \frac{1-\cos 2x}{2} dx = \frac{1}{2}\int 1 - \cos 2x \, dx = \frac{1}{2}\left(x - \frac{1}{2}\sin 2x\right) + C$.

$\sin^2 x \, dx = \int \frac{1-\cos 2x}{2} dx$

$= \frac{1}{2}\int 1 - \cos 2x \, dx$

$= \frac{1}{2}\left(x - \frac{\sin 2x}{2}\right) + C$

(b) $\int \cos^2 x \, dx = \int \frac{1+\cos 2x}{2} dx = \frac{1}{2}\int 1 + \cos 2x \, dx = \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + C$.

$\int \cos^2 x \, dx = \int \frac{\cos 2x + 1}{2} dx$

$= \frac{1}{2}\int \cos 2x + 1 \, dx$

$= \frac{1}{2}\left[\frac{\sin 2x}{2} + x\right] + C$

$\sin 2x = 2\sin x \cos x$

$\cos 2A = 1 - 2\sin^2 A$

$\int \sin^2 x \cos^2 x \, dx$

$= \frac{1}{4}\int 4\sin^2 x \cos^2 x \, dx$

$= \frac{1}{4}\int \sin^2 2x \, dx = \frac{1}{4}\int \frac{1-\cos 4x}{2} dx$

$= \frac{1}{8}\int 1 - \cos 4x \, dx$

$= \frac{1}{8}\left(x - \frac{\sin 4x}{4}\right) + C$.

Example.

$\int \frac{2z\,dz}{\sqrt[3]{z^2+1}}$

$\int \frac{2z\,dz}{\sqrt[3]{z^2+1}} = \int 2z(z^2+1)^{-1/3} dz$

$u^3 = z^2 + 1$

$3u^2\, du = 2z\, dz$

$\int \frac{3u^2 du}{u}$

$= \int 3u \, du = \frac{3u^2}{2} + C$

$= \frac{3}{2}(z^2+1)^{2/3} + C$.

Let $u = z^2 + 1$. Then $\frac{du}{dz} = 2z$.

$$\int \frac{2z\,dz}{\sqrt[3]{z^2+1}} = \int u^{-\frac{1}{3}} du = \frac{u^{\frac{2}{3}}}{\frac{2}{3}} + C = \frac{3}{2}u^{\frac{2}{3}} + C = \frac{3}{2}(z^2+1)^{\frac{2}{3}} + C.$$

Definite Integral Substitution and the Area Between Curves

The Substitution Formula

Theorem. *Substitution in Definite Integrals*

If g' is continuous on the interval $[a, b]$ and f is continuous on the range of g, then

$$\int_a^b f(g(x))\, g'(x)dx = \int_{g(a)}^{g(b)} f(u)\, du.$$

Example.

Evaluate $\int_{-1}^{1} 3x^2\sqrt{x^3+1}\, dx$.

Solution. We transform the integral and evaluate the transformed integral with the transformed limits given.

Let $u = x^3 + 1$. Then $\frac{du}{dx} = 3x^2$, when $x = -1, u = 0$ and $x = 1, u = 2$.

$$\int_{-1}^{1} 3x^2\sqrt{x^3+1}\, dx = \int_0^2 \sqrt{u}\, du = \frac{2}{3}\left[u^{\frac{3}{2}}\right]_0^2 = \frac{2}{3}\left(2^{\frac{3}{2}} - 0^{\frac{2}{3}}\right)$$
$$= \frac{2}{3}(2\sqrt{2}) = \frac{4\sqrt{2}}{3}.$$

Example.

Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot\theta \csc^2\theta\, d\theta$.

Solution. Let $u = \cot\theta$. Then $\frac{du}{d\theta} = -\csc^2\theta$.

When $\theta = \frac{\pi}{4}, u = \cot\frac{\pi}{4} = 1$ and $\theta = \frac{\pi}{2}, u = \cot\frac{\pi}{2} = 0$.

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot\theta \csc^2\theta\, d\theta = \int_1^0 u(-du) = \int_0^1 u\, du = \frac{u^2}{2}|_0^1 = \frac{1}{2}.$$

Example.

Evaluate $\int_0^{\frac{\pi}{2}} \frac{2\sin x\cos x}{(1+\sin^2 x)^3}\,dx$.

$u = 1+\sin^2 x$

$du = 2\sin x\cos x\,dx$.

$x=0,\ u = 1+\sin^2 0 = 1$

$x=\frac{\pi}{2},\ u = 1+\sin^2\frac{\pi}{2} = 2$.

$= \int_1^2 \frac{du}{u^3}$

$= \int_1^2 u^{-3}\,du$

$= \left[\frac{u^{-2}}{-2}\right]_1^2 = -\frac{1}{2}\left(\frac{1}{u^2}\right)_1^2 = -\frac{1}{2}\left(\frac{1}{4}-1\right)$

$= -\frac{1}{2}\left(-\frac{3}{4}\right) = \frac{3}{8}$ ✱

Definite Integrals of Symmetric Functions.

$f(-x) = -f(x)$ odd

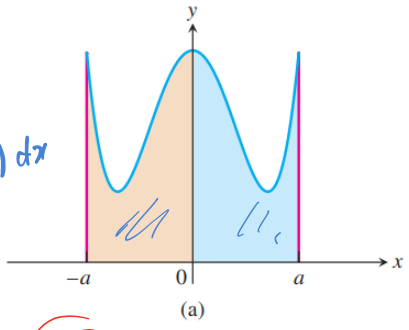
$f(-x) = f(x)$ even

$f(x)$ even

$\int_{-a}^{a} f(x)\,dx$

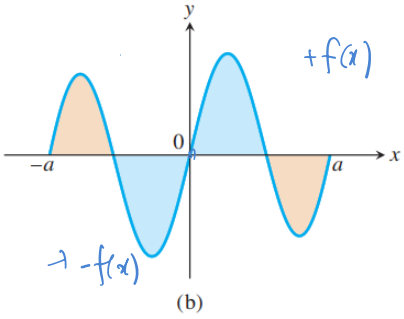
$= \int_{-a}^{0} f(x)\,dx + \int_0^a f(x)\,dx$

$= 2\int_0^a f(x)\,dx$



(a)

(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$



(b)

(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

$\int_{-a}^{a} f(x)\,dx$

$= \int_{-a}^{0} f(x)\,dx + \int_0^a f(x)\,dx$

$= 0$

Theorem.

Let f be continuous on the symmetric interval $[-a, a]$.

(a) if f is even, then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

(b) if f is odd, then $\int_{-a}^{a} f(x)\,dx = 0$.

even

$\int_{-2}^{2} (x^4 - 4x^2 + 6)\,dx$

$= 2\int_0^2 x^4 - 4x^2 + 6\,dx = 2\left[\frac{x^5}{5} - \frac{4x^3}{3} + 6x\right]_0^2$

$= 2\left(\frac{2^5}{5} - \frac{4(2^3)}{3} - 6(2)\right)$

$= 2\left(\frac{32}{5} - \frac{4}{3}(8) - 12\right)$

$= \frac{232}{15}$.

Example. *Integral of an Even Function*

Evaluate $\int_{-2}^{2}(x^4 - 4x^2 + 6)\,dx$.

Solution. $f(x) = x^4 - 4x^2 + 6$ satisfied $f(-x) = f(x)$. It is an even function and symmetric in interval $[-2,2]$, so

$$\int_{-2}^{2} (x^4 - 4x^2 + 6)\,dx = 2\int_0^2 (x^4 - 4x^2 + 6)\,dx$$

$$= 2\left[\frac{x^5}{5} - \frac{4x^3}{3} + 6x\right]_0^2 = 2\left(\frac{32}{5} - \frac{4(8)}{3} + 6(2)\right) = \frac{232}{15}.$$

odd

$\int_{-1}^{1} f(x)\,dx = 0$.

odd

$$\int_{-1}^{1} \frac{\tan x}{(1+x^2+x^4)}\,dx = 0.$$

Example.

Evaluate $\int_{-1}^{1} \frac{\tan x}{1+x^2+x^4}\,dx$.

Solution.

Let $f(x) = \frac{\tan x}{1+x^2+x^4}$. Since $f(-x) = -f(x)$, it is an odd function and so

$$\int_{-1}^{1} \frac{\tan x}{1+x^2+x^4}\,dx = 0.$$

Area Between Curves

Suppose we want to find the area of a region that is bounded above by the curve $y = f(x)$, below by the curve $y = g(x)$, and on the left and right by the lines $x = a$ and $x = b$.

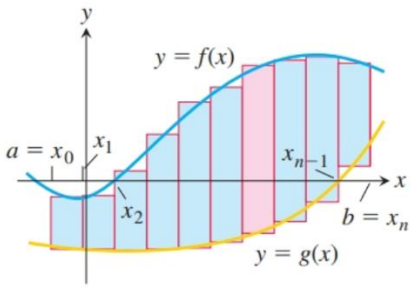


FIGURE 5.26 We approximate the region with rectangles perpendicular to the x-axis.

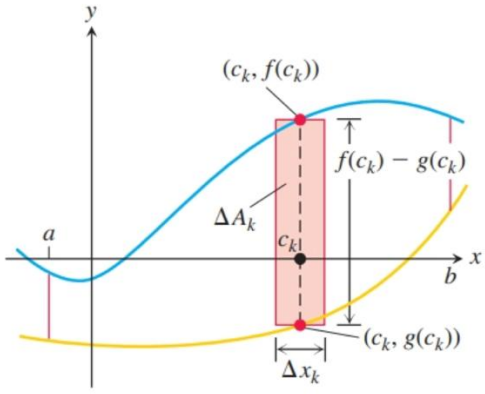


FIGURE 5.27 The area ΔA_k of the kth rectangle is the product of its height, $f(c_k) - g(c_k)$, and its width, Δx_k.

To see what the integral should be, we first approximate the region with n vertical rectangles based on a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$.

The area of the $k$the rectangle is

$$\Delta A_k = \text{height x width} = [f(c_k) - g(c_k)]\Delta x_k.$$

We approximate the area of the region by adding the areas of n rectangles.

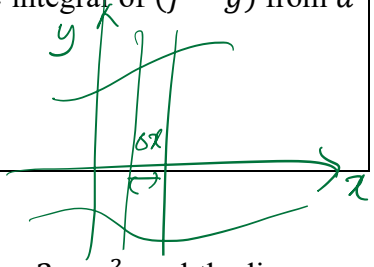
$$A \approx \sum_{k=1}^{n} \Delta A_k$$
$$= \sum_{k=1}^{n} [f(c_k) - g(c_k)]\Delta x_k.$$

As $||P|| \to 0$, the sums on the right approach the limit $\int_a^b [f(x) - g(x)]\,dx$ because f and g are continuous.

$$A = \lim_{||P||\to 0} \sum_{k=1}^{n} [f(c_k) - g(c_k)]\Delta x_k$$
$$= \int_a^b [f(x) - g(x)]\,dx.$$

Definition. If f and g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then the area of region between the curves $y = f(x), y = g(x)$, from a to b is the integral of $(f - g)$ from a to b:

$$A = \int_a^b [f(x) - g(x)]\,dx.$$



Example. Find the area of the region enclosed by the parabola $y = 2 - x^2$, and the line $y = -x$.

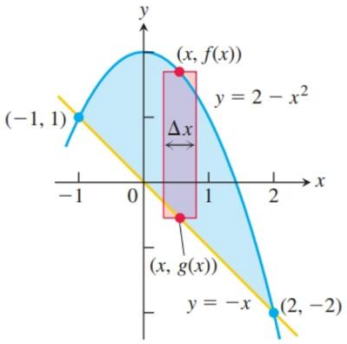
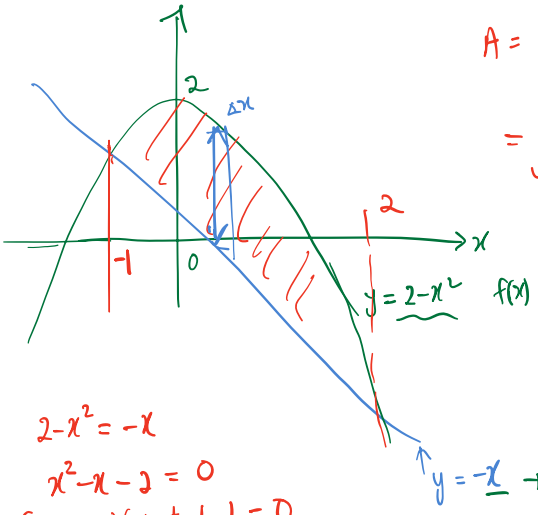


FIGURE 5.28 The region in Example 5 with a typical approximating rectangle from a Riemann sum.



$$A = \int_{-1}^{2} (2-x^2) - (-x)\,dx$$

$$= \int_{-1}^{2} 2 - x^2 + x\,dx$$

$$= \left[2x - \frac{x^3}{3} + \frac{x^2}{2}\right]_{-1}^{2}$$

$$=$$ _____ .

$2 - x^2 = -x$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$x = -1, \quad x = 2$

Example. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

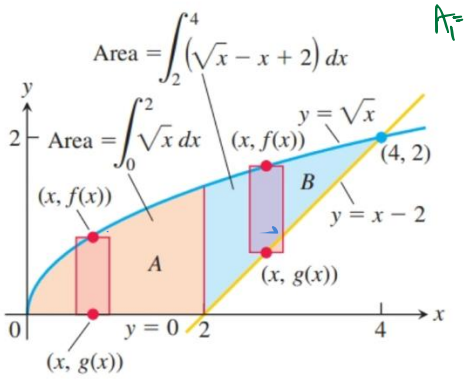
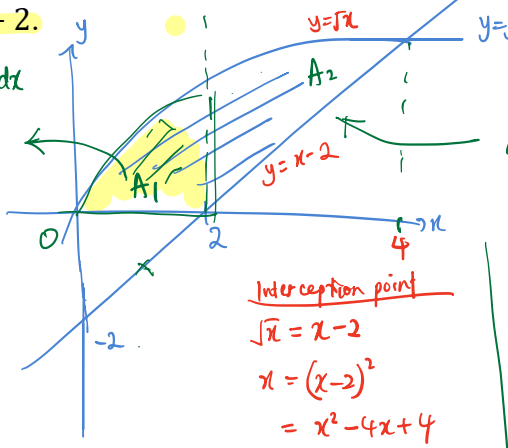


FIGURE 5.29 When the formula for a bounding curve changes, the area integral changes to become the sum of integrals to match, one integral for each of the shaded regions shown here for Example 6.



$$A_1 = \int_0^2 \sqrt{x}\,dx$$

$$A_2 = \int_2^5 \sqrt{x} - (x-2)\,dx$$

Interception point

$\sqrt{x} = x - 2$

$x = (x-2)^2$

$= x^2 - 4x + 4$

$x^2 - 5x + 4 = 0$

$(x-4)(x-1) = 0$

$x = 1, \quad x = 4.$

$$A_1 = \int_0^2 \sqrt{x}\,dx$$

$$= \frac{x^{3/2}}{3/2}\Big]_0^2$$

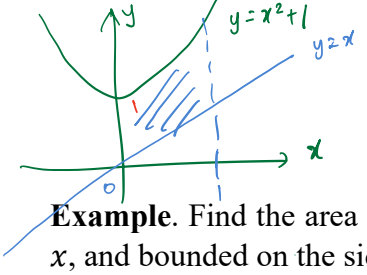
$$= \frac{2}{3}\left(2^{3/2}\right)$$

$$A_2 = \int_2^5 \sqrt{x} - x + 2\,dx$$

$$= \left[\frac{x^{3/2}}{3/2} - \frac{x^2}{2} + 2x\right]_2^5$$

$$=$$

$A = \frac{10}{3}$ unit²



$A = \int_0^1 (x^2+1) - x \, dx$

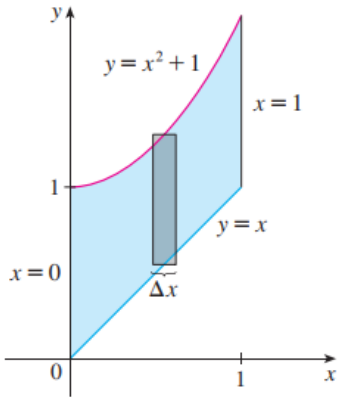
$= \int_0^1 x^2 + 1 - x \, dx$

$= \left[\frac{x^3}{3} + x - \frac{x^2}{2}\right]_0^1 =$ _____

Example. Find the area of the region bounded above by $y = x^2 + 1$, bounded below by $y = x$, and bounded on the sides by $x = 0$ and $x = 1$.

Solution. The upper boundary curve is $y = x^2 + 1$ and lower boundary curve is $y = x$. So we use the area formula with $f(x) = x^2 + 1, g(x) = x, a = 0$ and $b = 1$:

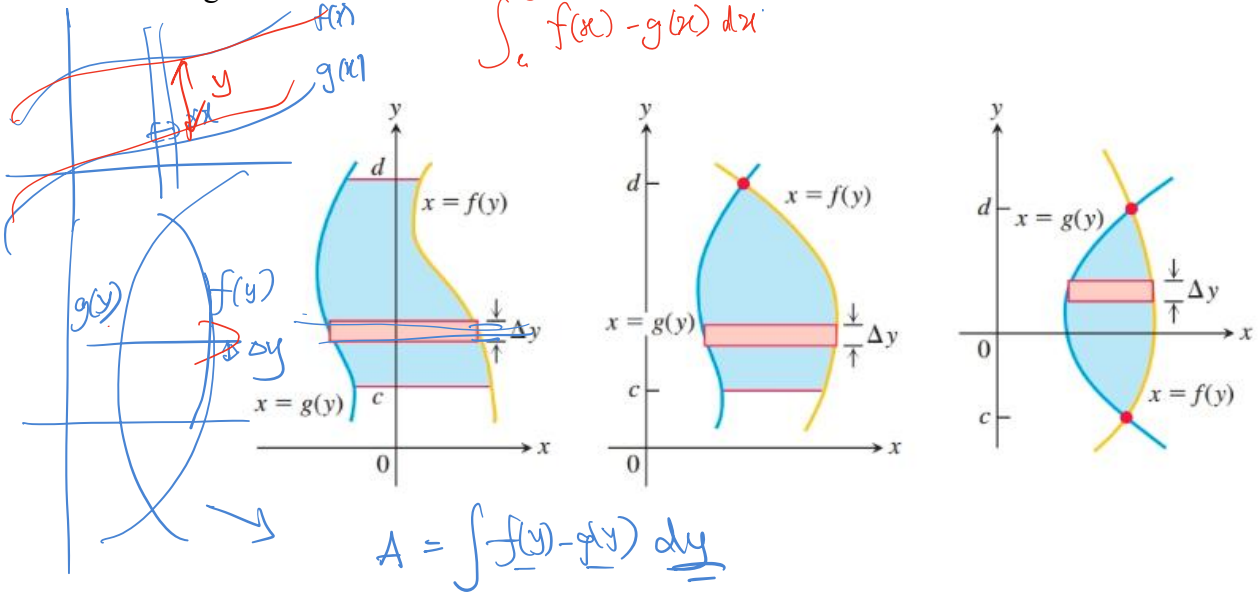
$$A = \int_0^1 [(x^2 + 1) - x]\, dx$$
$$= \int_0^1 (x^2 - x + 1)\, dx$$
$$= \left[\frac{x}{3} - \frac{x^2}{2} + x\right]_0^1$$
$$= \frac{5}{6}.$$



Integration with Respect to y

If a region's bounding curves are described by functions of y, the approximating rectangles are horizontal instead of vertical, and the basic formula has y in place of x. To find the areas of regions like these

$\int_a^b f(x) - g(x)\, dx$



$A = \int f(y) - g(y)\, dy$

Use the formula

$$A = \int_c^d [f(y) - g(y)]\, dy.$$

In this equation f always denotes the right-hand curve and g the left-hand curve, so $f(y) - g(y)$ is nonnegative.

Example. Find the area of the region in *Example* by integrating with respect to y.

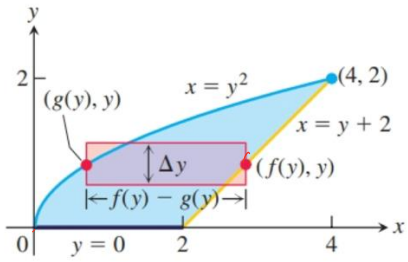


FIGURE 5.30 It takes two integrations to find the area of this region if we integrate with respect to x. It takes only one if we integrate with respect to y (Example 7).

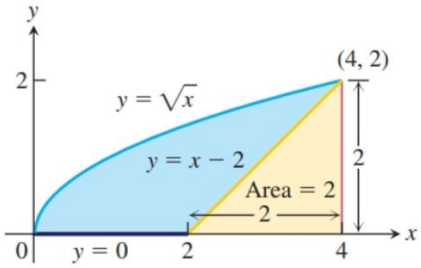
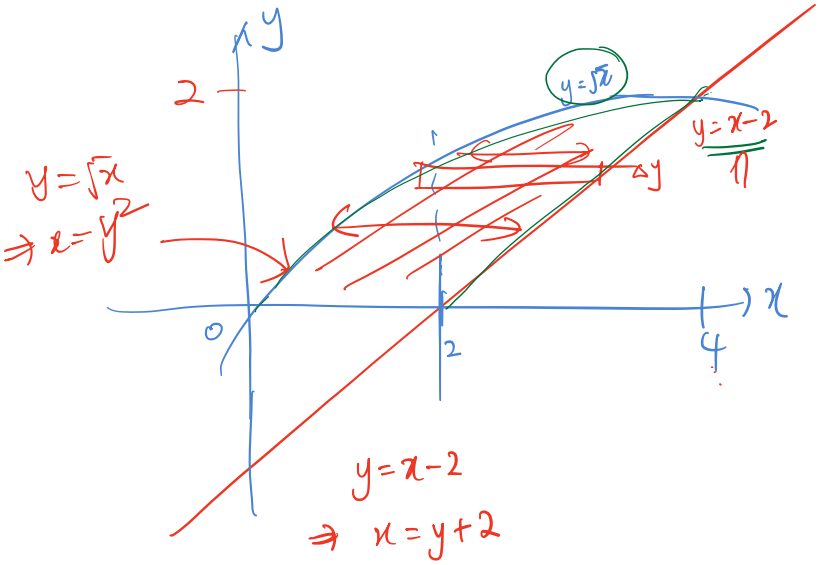


FIGURE 5.31 The area of the blue region is the area under the parabola $y = \sqrt{x}$ minus the area of the triangle.



$$\downarrow f(y) - g(y)$$

$$\int_0^2 (y+2) - y^2 \, dy$$

$$= \int_0^2 y + 2 - y^2 \, dy$$

$$= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_0^2$$

$$= \vdots$$

$$= \frac{10}{3}$$

Example. Find the area of the region bounded by the line $y = 2 - x$ and above by the curve $y = \sqrt{2x - x^2}$.

Solution.

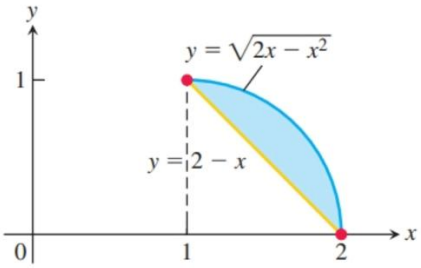


FIGURE 5.32 The region described by the curves in Example 7.

$y = 2-x$, $y = \sqrt{2x-x^2}$

$y^2 = 2x - x^2$

$y^2 + x^2 - 2x = 0$

$y^2 + (x-1)^2 - 1 = 0$

$y^2 + (x-1)^2 = 1$

circle

radius = 1

centre = (1, 0)

$(x-1)^2 = 1 - y^2$

$y = \sqrt{2x-x^2}$

$x - 1 = \sqrt{1-y^2}$

$x = 1 + \sqrt{1-y^2}$

$y = 2-x$

$A = \int_1^2 \sqrt{2x-x^2} - (2-x)\, dx$

$\int \sqrt{2x-x^2}\, dx$?

$A = \int_0^1 \left(1+\sqrt{1-y^2}\right) - (2-y)\, dy$

$A = \frac{1}{4}\left(\pi(1)^2\right) - \frac{1}{2}(1)(1)$

$= \frac{\pi}{4} - \frac{1}{2}$.

$x^2 + y^2 + xy = \tan(xy)$

$= \sec^2(xy)\left[x\frac{dy}{dx} + y\right]$

$\frac{d}{dx}(f(u)) = f'(u)\frac{du}{dx}$